

Indexical Shift and its interactions with Perspective Sensitive Items (PSIs): Indexical PSIs in Turkish

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Abstract The link between indexical shift and perspective shift is not well-understood. In indexical shift, indexicals nominals (e.g. *I, you, here, tomorrow*) are interpreted relative to a reported context, rather than the actual context of utterance. In perspective shift, perspective sensitive items (PSIs) (e.g. *tasty, local, foreigner*), are interpreted from the viewpoint of an individual other than the speaker. This study examines how PSIs and indexical shift interact in Turkish attitude reports, and shows that PSIs do not exhibit a uniform behavior. In light of novel data, this study shows that some PSIs, termed *Indexical PSIs* (e.g. motion verb *gel* ‘to come’, evidential markers), show shift together patterns with indexical nominals, being targeted by indexical shift. In other words, Indexical PSIs must shift when they appear in an indexical shifting context, and cannot shift without indexical shift. Other types of PSIs, on the other hand, do not interact with the occurrence of indexical shift. The existence of Indexical PSIs supports context over-writing approaches to indexical shift over alternative accounts that try to reduce indexical shift into simple feature deletion/impoverishment.

Keywords indexical shift, perspective shift, perspective sensitive items, Turkish, attitude reports

1 Introduction

Indexical elements such as English *I, you, here* and *yesterday* get their interpretations relative to the context in which they are uttered (Kaplan, 1977). For example, if (1) is uttered by *John* in *Los Angeles*, it means that John was born in Los Angeles, but if it is uttered by *Mary* in *Boston* it means that Mary was born in Boston.

(1) **I** was born **here**.

Indexicals are expressions that are used to refer to the referents of the speech-act coordinates like *author, addressee, location* and *time* (e.g. Kaplan, 1977; Schlenker, 2003; Anand and Nevins, 2004; Deal, 2020). For example, *I* is used to refer to the *author* (speaker) of the utterance, while *here* is used to refer to the *location* where the utterance was made, and thus sentence (1) means different things if uttered by different people and/or in different locations. However, there is a relatively large literature reporting that some languages, including Amharic (Schlenker, 1999), Zazaki (Anand and Nevins, 2004), Slave (Anand and Nevins, 2004), Uyghur (Shklovsky and Sudo, 2014), Nez Perce (Deal, 2020), and Turkish (Şener and Şener, 2011), allow indexical shift, a situation where an indexical element gets its referent from the reported context, rather than the actual context of utterance. For example, the Uyghur 1st person indexical *men* ‘I’ and the 2nd person indexical *sen* ‘you’ in (2) refer to the speaker and the addressee of the reported context, *Ahmet* and *Aygül*, rather than the actual speaker and addressee.

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- (2) Ahmet Aygül-ge [**men seni** yaxshi kör-ymen] di-di.
 Ahmet Aygül-DAT 1SG 2SG.ACC well see-IMPF.1SG say-PST
 ‘Ahmet_i told Aygül_j that he_i likes her_j.’
 (Uyghur, from Shklovsky and Sudo, 2014, pp.395)

Previous work suggests that indexical shift replaces the *author* and *addressee* of the embedded clause with the speaker and the addressee of the reported context (e.g. Anand and Nevins, 2004; Sudo, 2012; Shklovsky and Sudo, 2014; Deal, 2020), and thus in a sentence like (2) the 1st person indexical *men* picks the attitude holder *Ahmet* as its referent while the 2nd person indexical *sen* refers to his addressee *Aygül*.

A similar kind of shifty behavior is observed with *perspective sensitive items* (PSIs), whose interpretation can be context dependent (e.g. epithets like *idiot*, predicates of personal taste like *tasty*, relative socio-cultural expressions like *foreigner*). PSIs are typically interpreted from the point of view of a *perspective holder*. For example, to be judged *true* or *false*, the sentence in (3) needs to be interpreted relative to the perspective of a person (e.g. Speas and Tenny, 2003; Lasersohn, 2005; Oshima, 2006; Stephenson, 2007; Bylinina et al., 2014; Anand and Korotkova, 2021).

- (3) John is a **foreigner**.

In other words, the truth of (3) depends on where the perspective holder, the person under whose perspective *foreigner* is evaluated, is from. As Bylinina et al. (2014) put it, the sentence in (3) roughly means ‘John is a foreigner to the perspective holder’s country’. It is generally assumed that the perspective holder of a clause is represented with the *judge* coordinate of the context (Lasersohn, 2005; Stephenson, 2007, and subsequent work). In simple sentences like (3) the perspective holder is the speaker of the sentence by default (e.g. Lasersohn, 2005; Bylinina et al., 2014; Anderson, 2021; Kaiser, 2022a). However, in some syntactic contexts like attitude reports, *judge* shifts to a different person than the speaker, the phenomenon known as *perspective shift*. When perspective shift occurs, the PSIs are interpreted under the perspective of a new perspective holder, rather than the perspective of the speaker. (4) illustrates that *perspective shift* can optionally occur in English attitude contexts embedded under the verb *to think*.

- (4) Mary thinks that John is a **foreigner**.
 a. non-shifted → ‘John is from a country other than where speaker is from’
 b. shifted → ‘John is from a country other than where John is from’
 (adapted from Bylinina et al., 2014)

Without perspective shift, the relative socio-cultural expression *foreigner* is interpreted under the perspective of the speaker (4-a). When perspective shift takes place, *foreigner* is interpreted under the perspective of the attitude holder *Mary* (4-b).

At least at first glance, *indexical shift* and *perspective shift* appear to share some common properties. For instance, they both replace one of the context parameters (*author* or *judge*) with the attitude holder, and they can take place in same syntactic environments (e.g. attitude reports). Despite these similarities, the relatively large literatures on these phenomena seem mostly disconnected, and the relation between *indexical shift* and *perspective shift* is not yet fully understood. Considering that *author* and *judge* coincide to refer to the speaker of the sentence in non-shifted (indexical or perspective) clauses, it might be the case that the *judge* coordinate is sensitive to the value of the *author* coordinate, and it refers to the same entity with *author* by default. In that case, indexical shift might indirectly change the interpretations of PSIs by changing the referent of the *author* coordinate (i.e. indexical shift might force perspective shift). Alternatively, if *author* and *judge* coordinates are totally independent from each other, indexical shift and perspective shift might independently change the referents of the *author* and *judge* coordinates without affecting the other. It is also possible that there might be limitations as to how many ‘shifting’ operations can take place, and the occurrence of indexical shift might block the occurrence of perspective shift (or vice versa).

Recent work investigates these possible interactions between indexical shift and perspective shift, and suggests that these two phenomena are in a subset-superset relation, where availability of indexical

shift entails availability of perspective shift (but not vice-versa) (Bylinina et al., 2014; Sundaresan, 2021). Sundaresan (2021) suggests that indexical shift and perspective shift can be unified under a more general model of *attitude shift*, where various types of shifty elements are ordered in a single implicational continuum of attitude shift. According to this view, if indexical shift is possible in a linguistic environment in a language, perspective shift must also be available in the same environment.

In the current study we investigate the interactions between indexical shift and perspective shift from a different angle, testing if the two can (or must) occur at the same time in Turkish attitude reports, where indexical shift is optionally available ¹. We investigate how six different types of perspective sensitive items (PSIs) behave, and show that PSIs do not exhibit a uniform behavior and can be categorized into two main types depending on their interactions with indexical shift, as summarized in the table in (6).

- (5) Burak_i **ben**_{speaker/i} Ankara-ya git-ti-m san-ıyor.
 Burak 1SG Ankara-DAT go-PST-1SG think-PROG
 ‘Burak thinks that {I/he} went to Ankara.’

- (6) Shift together patterns between indexicals and different types of PSIs²

Group	PSI type	Shift together	Relevant coordinate
Indexical PSIs	Motion verb <i>gel</i> ‘to come’ Evidentiality	YES	<i>author</i>
Non-indexical PSIs	Epistemic modals Predicates of personal taste Relative expressions Epithets	NO	<i>judge</i>

Some PSIs, which we term *Indexical PSIs*, show indexical-like properties and exhibit shift together patterns with the 1st person indexicals. In other words, Indexical PSIs (e.g. motion verb *gel* ‘to come’ and evidential markers) must shift when indexical shift takes place, and cannot shift when indexical shift does not take place. Other PSIs, which we call *non-indexical PSIs*, on the other hand, are not affected by the (non-)occurrence of indexical shift and exhibit their own patterns (e.g. optionally shifting, not being able to shift) regardless of indexical shift. The behaviors of these PSIs are not affected by occurring in an indexical shifting environment.

These results have important implications regarding the theoretical mechanisms of how PSIs are interpreted and what type of elements can be targeted by the indexical shift operation. First, our findings show that PSIs are not uniform in how their interpretations are derived: Only a subset of PSIs (Indexical PSIs) are sensitive to the *author* coordinate, while most others are not directly sensitive to this parameter. This suggests PSIs might be interpreted relative to different context coordinates like *judge* and *author*. Second, our findings show for the first time that indexical shift can target not only nominal indexical elements (e.g. *I*, *you*) but also predicates (e.g. *gel* ‘to come’) and modals (evidential markers), which suggests that these elements can have indexical properties. This supports the context-overwriting approaches to indexical shift (e.g. Anand and Nevins, 2004; Deal, 2020), and are not directly predicted by approaches that reduce indexical shift into semantic/syntactic binding (e.g. Podobryaev, 2014) or simple feature deletion/impoverishment operations (e.g. Messick, 2023).

To account for the indexical-like properties of indexical PSIs (e.g. motion verb *gel* ‘to come’ and evidential marker ‘-*mİş*’), I propose that these PSIs have semantic definitions along the lines of (7), which makes them sensitive to the *author* coordinate.

- (7) a. *gel* = an event of movement towards the *author*
 b. ‘-*mİş*’ = the *author* lacks direct evidence for reported proposition

¹ Unless cited otherwise, all Turkish data in this paper comes from the judgments of the author, who is a native speaker of Turkish. Most examples have been surveyed among 15 other native speakers. Except for some examples, where inter-speaker variation exists, all native speakers confirmed their judgments for the data provided.

² Though the list of PSIs can be extended with more different types, the current study is limited to six different types of PSIs for time and space reasons: (i) motion verbs, (ii) evidentiality, (iii) epistemic modals, (iv) epithets (v) predicates of personal taste, and (vi) socio-cultural expressions. Future research should extend this list to cover other types of PSIs.

Assuming definitions along the lines of (7), I follow previous work in Turkish (Özyıldız et al., 2018; Akkuş, 2019; Oğuz et al., 2020) and suggest that a context-overwriting shifty operator (OP_{AUTH}) can optionally merge the periphery of the finite embedded clauses in Turkish and replace the *author* coordinate of the context with the attitude holder, as schematized in (8).

- (8) Speaker = *Dursun_i*
- a. [TP Burak_k [CP/TP 1SG_i ... *come_i/evidentials_i*] think] non-shifted
 - b. [TP Burak_k [CP/TP OP_{AUTH} [1SG_k ... *come_k/evidentials_k*]] think] shifted

Once this OP_{AUTH} merges, it shifts the readings of all items that are sensitive to the *author* coordinate like 1st person pronouns and indexical PSIs.

Lastly, I use *indexical PSIs* as novel diagnostics to identify indexical shift and show for the first time that indexical shift can take place inside finite embedded clauses with accusative subjects like (9), where embedded verb agreement is optional. Based on the fact that the accusative marked 1st person pronoun in (9) cannot have shifted interpretations, previous work suggested that indexical shift may not be available in sentences where embedded subject is marked with accusative (e.g. Oğuz et al., 2020; Başoğlu, 2021). However, as pointed out by (?), it is still unclear if the accusative marked subject is not able to shift because of its syntactic location, which is assumed to be higher than the indexical shifting OP_{AUTH} , or simply because the OP_{AUTH} does not merge in the structure at all.

- (9) Burak_i **ben-i**_{speaker/*i} Ankara-ya git-ti-(m) san-ıyor.
 Burak 1SG-ACC Ankara-DAT go-PST-1SG think-PROG
 ‘Burak thinks that {I/*he} went to Ankara.’

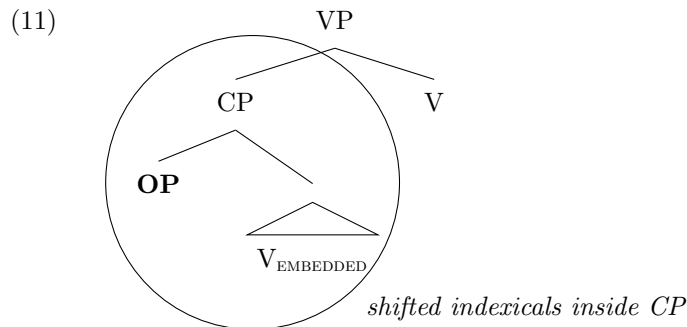
In this study, I show that the OP_{AUTH} can indeed merge into the embedded clause of sentences with accusative subjects as (9), and shift all indexical items (including indexical PSIs) other than the accusative-marked subject. Moreover, a careful investigation reveals that the optionality of embedded verb agreement is not random or coincidental, but signals the (non-)occurrence of indexical shift in the embedded clause. Precisely, the lack of a full/matching agreement marker on the embedded verb signals the merge of the indexical shifting OP_{AUTH} . Considering that the accusative-marked DP (*ben-i* ‘I-ACC’ in (9)) stays non-shifted even when indexical shift takes, I suggest that the accusative DP is located higher than the position occupied by the OP_{AUTH} and thus stays outside the scope of the operator as schematized in (10).

- (10) Speaker = *Dursun_i*
- a. [TP Burak_k [CP/TP 1SG_i ... *come_i/evidentials_i*] think] non-shifted
 - b. [TP Burak_k [^{sc}p/TP OP_{AUTH} [1SG_k ... *come_k/evidentials_k*]] think] shifted

The paper is organized as follows: Section 2 provides background on indexical shift and summarizes previous work on indexical shift in Turkish. Section 3 provides novel data demonstrating that some PSIs follow shift together patterns with indexicals while others do not, and lays out the analysis developed in this paper. Section 4 provides novel data with Accusative Subject Clauses (ASCs) in Turkish and extends the proposed analysis to account for indexical shift in ASCs. Section 5 concludes the paper.

2 Background

One generally employed analysis of indexical shift is the shifty operator (or the monster operator) approach, originally proposed by Anand and Nevins (2004) and developed by subsequent work (e.g. Sudo, 2012; Shklovsky and Sudo, 2014; Deal, 2020, but see Schlenker (1999; 2003) for a different approach). Under this view, a context-overwriting operator (OP) merges in the left periphery of the embedded clause (11), and forces all indexicals in its scope to be interpreted inside the reported context, rather than the actual speech context. For the Uyghur example in (2), this means that the OP merges in the left periphery of the embedded clause and the 1st person indexical refers to *Ahmet* and the 2nd person indexical refers to *Aygül*, who are the *author* and the *addressee* of the reported context, respectively.



Anand and Nevins (2004) report that all indexicals in an embedded clause must be interpreted relative to the same context (either shifted or non-shifted), introducing the well-known *Shift Together* constraint. Deal (2020) provides cross-linguistic data and shows that not all indexicals shift together, but the indexicals that depend on the same context coordinate (e.g. *author*, *addressee*, *location*) must shift together. She proposes that each context parameter can be shifted by a separate context-overwriting operator. For example, she argues that the OP_{AUTHOR} is responsible for shifting the *author* parameter of the context, while $OP_{ADDRESSEE}$ is responsible for shifting the addressee parameter. Considering that the interpretation of all 1st person indexicals depends on the *author* parameter, all 1st person indexicals must have shifted interpretations when they are under the scope of OP_{AUTHOR} . Crucially, under this view, the merge of OP_{AUTHOR} does not affect the interpretation of any other types of indexicals like 2nd person and *locatives*, since their interpretations are dependent on the *addressee* and *location* parameters of the context, respectively. Considering this, it is possible for a language to allow 1st person indexicals to have shifted interpretations, while disallowing shifted interpretations of 2nd person indexicals (such as *Slave*, as reported in Deal (2020)). To account for the cross-linguistic variation, Deal (2020) proposes that at least four different types of context-overwriting operators must exist: OP_{AUTHOR} for 1st person indexicals like *I*, $OP_{ADDRESSEE}$ for 2nd person indexicals like *you*, $OP_{LOCATIVE}$ for locative indexicals like *here*, and OP_{TIME} for temporal indexicals like *tomorrow*.

Turkish allows indexical shift in finite embedded clauses like (12), which has received some attention in the literature (Şener and Şener, 2011; Özyıldız, 2012; Özyıldız et al., 2018; Akkuş, 2019; Korotkova, 2019; Kuram, 2020; Oğuz et al., 2020).

- (12) Burak_i (**ben**)_{speaker/i} Ankara-ya git-ti-m san-ıyor.
 Burak 1SG Ankara-DAT go-PST-1SG think-PROG
 ‘Burak thinks that {I/he} went to Ankara.’

The 1st person indexical *ben* in the embedded clause of (12) can be interpreted as referring to the speaker (non-shifted) or the attitude holder *Burak* (shifted). In Turkish, indexical shift is observed with various types of indexicals such as the 1st person *ben*, 2nd person *sen* ‘you’, locative *bura* ‘here’, and temporal *yarın* ‘tomorrow’, embedded under attitude verbs like *de* ‘to say’, *san* ‘to think’, *iste* ‘to want’, and *bil* ‘to know’ (Özyıldız, 2012; Oğuz et al., 2020).

It is important to establish the fact that these are true instance of indexical shift, rather than quotation (e.g. Burak thinks, ‘I went to Ankara’). Previous work uses diagnostics like NPI licensing and *wh*-licensing to show that embedded material in sentences like (12) is not direct quotation (e.g. Özyıldız, 2012; Akkuş, 2019; Oğuz et al., 2020). In affirmative sentences, the Turkish NPI *kimse* ‘anyone’ must be licensed by a negative item that takes scope over it (Kelepir, 2001). As in (13), the NPI *kimse* ‘anyone’ in the embedded object position can be licensed by embedded (13-a) or matrix (13-b) negation. Crucially, the sentences are ungrammatical in the absence of negation.

- (13) a. Ali Burak **kimse-yi** gör-*(**me**)-di san-ıyor.
 Ali Burak anyone-ACC see-NEG-PST think-PROG
 ‘Ali thinks that Burak didn’t see anyone.’
 b. Ali Burak **kimse-yi** gör-dü san-*(**mı**)-yor
 Ali Burak anyone-ACC see-PST think-NEG-PROG
 ‘Ali doesn’t think that Burak saw anyone.’

Considering that quotation creates an opaque domain, blocking the licensing of NPI by a negative item from outside (14), examples like (15) with shifted indexicals show that these are instances of indexical shift, but not direct quotation. If the embedded clause of (15) was a direct quotation, this would prevent the licensing of the NPI item *kimse* and lead to ungrammaticality, which is not the case.

(14) *John did **not** say “I saw **anyone**”.

(15) Burak_i (ben)_i **kimse-yi** gör-dü-m san-mı-yor.
 Burak1SG anyone-ACC see-PST-1SG think-NEG-PROG
 ‘Burak doesn’t think that {I/he} saw anyone.’

Further evidence for that embedded material in those clauses are not direct quotation comes from *wh*-licensing facts. Considering that direct quotations are islands for *wh*-extractions, as in (16), sentences like (17) suggest that examples of indexical shift are not direct quotations.

(16) ***What**_{*i*} did the teacher say “I assigned a lot of *t_i*”.

(17) Burak_i (ben)_i **kim-i** gör-dü-m san-mı-yor?
 Burak1SG who-ACC see-PST-1SG think-NEG-PROG
 ‘Who does Burak think that {I/he} saw?’

Turkish does not have overt *wh*-movement, so the example in (17) does not illustrate an overt extraction of the *wh*-element. However, the fact that the *wh*-item *kim* ‘who’ can get matrix scope interpretation shows that it can covertly be extracted into the matrix domain, which suggests that the embedded clause in (17) is not direct quotation.

So, the NPI licensing and *wh*-licensing facts discussed above show that we are dealing with true instances of indexical shift. We mostly use affirmative sentences in the rest of the paper for demonstration purposes, but NPI-licensing and *wh*-movement diagnostics were used to confirm the non-quotational status of each data point provided.

In previous work, Şener and Şener (2011) report that the availability of shifted readings may depend on the phonological form of the indexical elements. For example, they suggest that shifted readings are available when the 1st person indexical is phonologically silent (*pro*), but are unavailable when the indexical is overt. Şener and Şener (2011) propose that only a sub-type of silent 1st person indexicals can shift, which they name *pro_{de se}*. They suggest that this specific pronoun comes with a context-overwriting operator OP (in the sense of Anand and Nevins (2004)) attached in its nominal structure (approximately left periphery of the NP layer), and this OP makes the indexical interpreted inside the reported context, rather than the speech context. Özyıldız et al. (2018) and Akkuş (2019)³ on the other hand, follow the more traditional version of the shifty operator analysis of Anand and Nevins (2004), and assume that a context-overwriting operator OP can attach in the left periphery of the embedded clauses in Turkish, making indexical shift available. Oğuz et al. (2020) mainly build on top of these analyses but report that data regarding indexical shift is more complicated. Precisely, they show that different types of indexicals (e.g. 1st person and temporal) show different properties under different embedding verbs. For example, they report that the embedding verb *de* ‘to say’ forces a shifted reading for person indexicals (e.g. 1st person and 2nd person) in the embedded clause, while *san* ‘to think’ only allows, but does not require, shifted readings. Moreover, some indexicals like *bura* ‘here’ shift under the embedding verb *de* ‘to say’, while they do not shift under the embedding verb *san* ‘to think’. Examples in (18) (adapted from Özyıldız (2012)) show how *bura* ‘here’ behaves differently under the embedding verbs *say* ‘to say’ and *san* ‘to think’.

(18) If uttered by *Dursun* in *Los Angeles*:

- a. Ali İzmir-de [**bura**-nın ne-ler-i güzel] de-miş-ti?
 Ali İzmir-LOC here-GEN what-PLU-3SG.POSS pretty say-DUB-PST
 ‘In İzmir, what did Ali say was pretty in {İzmir/Los Angeles}?’

³ Akkuş (2019) reports and analyzes a different dialect of Turkish, where violations of the *Shift Together* constraint of indexical shift is possible. He builds a more detailed analysis of indexical shift to account for these violation cases. None of our informants share the judgments reported in Akkuş (2019), so we do not provide details of his analysis in this paper.

- b. Ali İzmir-de [**bura**-nın ne-ler-i güzel] san-mış-tı?
 Ali İzmir-LOC here-GEN what-PLU-3SG.POSS pretty think-DUB-PST
 ‘In İzmir, what did Ali think was pretty in {*İzmir/Los Angeles}?’

While (18-a) with the embedding verb *de* ‘to say’ is ambiguous regarding how *bura* ‘here’ is interpreted (shifted or non-shifted), (18-b) with the embedding verb *san* ‘to think’ is unambiguous and *bura* ‘here’ can only be interpreted as referring to the location where sentence is uttered (non-shifted). This shows that the indexical *bura* ‘here’ can shift under the embedding verb *de* ‘to say’, but it cannot shift under the embedding verb *san* ‘to think’.

To account for these facts, Oğuz et al. (2020) adopt Deal’s (2020) approach and suggest that specific types of OPs can merge to the left periphery of the finite embedded clauses (to be selected by embedding verbs) and lead to shifted interpretations of indexicals of their own type. For example, under their account, the embedding verb *de* ‘to say’ in (18-a) can select for an embedded clause with an OP_{LOC} in its periphery, and as a result lead to shifted readings of locative indexicals like *bura* ‘here’, while the embedding verb *san* ‘to think’ cannot select for an embedded clause with the OP_{LOC} and thus the locative indexical *bura* ‘here’ cannot have shifted readings in (18-b). However, *san* ‘to think’ can select for an embedded clause with an OP_{AUTHOR} in its periphery, and as a result lead to shifted readings of 1st person indexicals, as in (12)⁴. As laid out in the previous section (8), this paper builds on this analysis and argues that indexical PSIs (e.g. motion verb *gel* ‘to come’ and evidential markers) are also targeted by the context shifting operator OP_{AUTHOR} .

3 Perspective sensitive elements inside *Indexical Shift* environments

This section provides novel data in order to investigate how Perspective Sensitive Items (PSIs) are interpreted inside indexical shifting environments. Novel data with PSIs like the motion verb *gel* ‘to come’, evidential markers, epistemic modals (e.g. *may*, *might*), predicates of personal taste (e.g. *tasty*, *fun*), relative socio-cultural expressions (e.g. *local*, *foreigner*), and epithets (e.g. *idiot*, *bastard*) shows that the PSIs in Turkish mostly behave differently from one another, forming at least four different groups based on their behaviors in attitude reports. Table in (19), repeated from (6), summarizes our findings, categorizing different types of PSIs into two main types based on their interactions with the 1st person indexicals.

(19) Shift together patterns between indexicals and different types of PSIs

Group	PSI type	Shift together	Relevant coordinate
Indexical PSIs	Motion verb <i>gel</i> ‘to come’ Evidentiality	YES	<i>author</i>
Non-indexical PSIs	Epistemic modals Predicates of personal taste Relative expressions Epithets	NO	<i>judge</i>

Crucially, *indexical PSIs* (e.g. motion verb *gel* ‘to come’, evidentials) show ‘shift together’ patterns with the 1st person indexicals and only shift when indexical shift takes place. *Non-indexical PSIs*, on the other hand, behave similarly in that they are not affected by the (non-)occurrence of indexical shift, even though they show unique patterns when it comes to their semantic details.

The following subsections provide relevant datapoints for each type of PSIs to demonstrate how they behave in indexical shifting environments.

⁴ Oğuz et al. (2020) propose that some form-specific OPs exist in the language, targeting only silent 1st person indexical *pro*.1SG ($OP_{AUTH-pro}$) or emphatic 1st person *ben kendim* ($OP_{AUTH-emphatic}$). These details are irrelevant for the purposes of the current study, and thus not discussed in further detail.

- (23) If *Dursun* is speaking:
 Burak Ali Ankara-ya *gel*-di san-iyor.
 Burak Ali Ankara-DAT come-PST think-PROG
 ‘Burak thinks that Ali came to Ankara.’
 a. → ‘Dursun (or addressee) is in Ankara.’ (non-shifted)
 b. → ‘Burak is in Ankara.’ (shifted)

Precisely, *gel* ‘to come’ in (23) can be interpreted as a movement towards the speaker *Dursun* (or the addressee) by default (23-a), or it can be interpreted as a movement towards the attitude holder *Burak* as a result of perspective shift (23-b). So, *gel* ‘to come’ can have shifted interpretations in finite embedded clauses, where indexical shift is also available (as previously shown in (12)).

Establishing this fact, the next step is to test whether indexical shift and perspective shift can (or must) take place at the same time, when a PSI (e.g. *gel* ‘to come’) and an indexical element occur in the same environment. This can be tested in sentences like (24), which has the 1st person indexical *ben* as the embedded subject, and the motion verb *gel* ‘to come’ as the embedded verb. As shown in Section 2, the 1st person indexical *ben* can be interpreted as referring to the speaker (non-shifted) or to the attitude holder (shifted). Also, as we saw above, the motion verb *gel* can be interpreted as a movement towards the speaker or the speaker’s addressee (non-shifted), or as a movement towards the attitude holder (shifted). Considering that there are two elements that each can have *shifted* or *non-shifted* readings, (24) can technically be ambiguous in four different ways, if the indexical *ben* and motion verb *gel* can shift independently from each other. This is not what we observe (24). Subscripts indicate who is in *Boston* (the destination of the movement denoted by *gel*).

- (24) If *Dursun* is speaking in *Los Angeles*
 Ev arkadaş-ım [CP (*ben*) Boston-a gel-ey-im] ist-iyor.
 House mate-1SG 1SG B.-DAT come-OPT-1SG want-PROG
 Lit: ‘My roommate wants me to come to Boston’
 a. neither shifts → ‘roommate wants **speaker** to go to Boston_{#spkr/addr}.’
 b. #both shift → ‘roommate wants **roommate** to go to Boston_{roommate}.’
 c. *only *gel* shifts → ‘roommate wants **speaker** to go to Boston_{roommate}.’
 d. *only *ben* shifts → ‘roommate wants **roommate** to go to Boston_{#spkr/addr}.’

Out of the logically possible four interpretations, (24) can have only two, one of which is odd/nonsensical. The first available reading is where neither the indexical *ben* nor the motion verb *gel* shifts (24-a). In this reading, the indexical *ben* refers to the speaker, and *gel* describes a motion towards the addressee of the speaker (via listener-anchoring), which is visible with the implication that the addressee of the sentence should be in Boston⁶. This yields a reading like ‘The roommate wants the speaker to move towards the addressee’. The second grammatically available reading is where both indexical *ben* and the motion verb *gel* shift (24-b). In this reading, the indexical *ben* refers to the attitude holder *the roommate*, and *gel* also describes a motion towards *the roommate*, yielding to a reading like ‘The roommate wants the roommate to move towards the roommate’, which is odd/nonsensical. Importantly, both interpretations require that the indexical *ben* and the PSI *gel* are get interpreted inside the same context: either the context of utterance (24-a), or the reported context (24-b). The readings where only one of these items shift (but not the other) are not available in the language (24-c)-(24-d), even in contexts/scenarios where those readings would be felicitous. For example, (24-c) is not available even in a context where ‘the roommate is in Boston, and the roommate wants the speaker to go to Boston’, and (24-d) is not available even in a context where ‘the speaker is in Boston, and the roommate wants to go to Boston’.

These facts suggest the motion verb *gel* ‘to come’ must undergo what we have so far called *perspective shift* when *indexical shift* takes place, and that it cannot undergo *perspective shift* when *indexical shift* doesn’t take place. In other words, we see that the motion verb *gel* ‘to come’ and the 1st person indexicals must ‘shift together’. As I explain below, this shift together behavior follows naturally from my analysis.

⁶ *Gel* ‘to come’ is not interpreted as a movement towards the speaker here, since the target of the movement is explicitly *Boston*, and we know from the context that the speaker is in Los Angeles. The crucial observation here is that the motion verb describes a movement towards the participant of the speech act context (i.e. *addressee*), but not towards a referent from the embedded context (e.g. *attitude holder*). Also, since the attitude context does not have an addressee (as a thought), the listener anchoring is not available in shifted readings.

If *gel* ‘to come’ is interpreted relative to the *author* coordinate of the context, just like the 1st person indexical, it must shift whenever indexical shift (of the 1st person) takes place and it cannot shift without indexical shift.

3.1.2 Evidentiality and Indexical Shift

Evidentiality encodes the source of information or strength of evidence for a reported proposition (Aikhenvald, 2004). While English uses various types of adverbs or modal verbs to encode evidentiality (e.g. *apparently*), other languages including (but not limited to) Bulgarian (Izvorski, 1997), Cusco Quechua (Faller, 2002), Korean (Park, 2014), Finnish (Kaiser, 2022b) St’at’incets (Matthewson et al., 2007), and Turkish (Şener, 2011) use specific evidential affixes or clitics. For example, Turkish uses evidential past tense morpheme ‘-*mİş*’ to roughly express the lack of direct evidence for the described situation (Göksel and Kerslake, 2005; Şener, 2011; Meriçli, 2016, a.o). The contrast between the regular past tense morpheme ‘-*DI*’ and the evidential past tense marker ‘-*mİş*’ is demonstrated below.

- (25) If *Dursun* is speaking:
- a. Ali şarkı söyle-**di**.
Ali song sing-PST
‘Ali sang a song.’ → ‘Dursun has direct evidence’
 - b. Ali şarkı söyle-**miş**.
Ali song sing-EVID
‘Ali sang a song.’ → ‘Dursun lacks direct evidence’

Sentences (25-a) and (25-b) are minimally different in that the regular past tense marker ‘-*DI*’ is used in (25-a) while the evidential past tense marker ‘-*mİş*’ is used in (25-b). If uttered by *Dursun*, (25-a) implies that *Dursun* has direct evidence for the reported event (e.g. he saw or heard Ali while Ali was singing). On the other hand, (25-b) implies that *Dursun* lacks direct evidence for the reported event, and he received the information from an external source (e.g. someone else told Dursun that Ali sang a song). Considering that evidentiality is perspective sensitive, being evaluated based on the evidence that author of an utterance has (Izvorski, 1997; Speas and Tenny, 2003; Stephenson, 2007, a.o.), it is natural/expected that (25-b) implies that the speaker, as the default perspective holder, lacks direct evidence for the reported situation.

Previous literature reports that the evidentials can also undergo *perspective shift* in attitude reports, and get evaluated under the perspective of the attitude holder (Sauerland and Schenner, 2007; Stephenson, 2007; Matthewson et al., 2007; Bylinina et al., 2014; Korotkova, 2015, 2019; Kaiser, 2024). This is observed in Turkish as well (Şener, 2011; Meriçli, 2016), as in (26), where the evidential marker ‘-*mİş*’ signals lack of direct evidence for the reported proposition.

- (26) Burak Ali öl-müş san-ıyor.
Burak Ali die-EVID think-PROG
‘Burak thinks Ali apparently died.’

Importantly, (26) indicates that the attitude holder *Burak* lacks direct evidence for the reported situation, and (26) is acceptable/felicitous even in contexts where the actual speaker has direct evidence for (or against) the reported situation.

Establishing these facts, we can test sentences that have both indexical elements and evidentials to check whether they can/must shift together. Understanding the interactions between indexicals and evidentiality requires building up some context scenarios, since interpretations of both of these items highly dependent on contextual information. We can use a context like (27) and a sentence like (28) as a baseline.

- (27) Context 1: *At an online store where real estate properties are sold, Dursun mistakenly clicks a button and buys a house without realizing that the click leads to the purchase of the house. A while later, he receives an email from the website, confirming the purchase he just made. He goes: ‘OMG, I apparently bought a house by mistake!’*
- (28) If *Dursun* is speaking:
 (Ben) yanlış-lık-la ev al-mış-ım!
 1SG wrong-NMLZ-INS house buy-evid-1SG
 ‘I apparently bought a house by mistake!’

Dursun can use (28) to describe the situation explained in Context 1. Importantly, Dursun does not realize that he bought a house at the time of the purchase, and he can use the evidential marker ‘-*mış*’ to report that his purchase was unintentional (e.g. he did not know that he made the purchase). Sentences like (28) are important in that they make use of an indexical element (i.e. 1st person *ben*) and a perspective sensitive element (i.e. *evidentiality*) at the same time.

A sentence like (28) can be embedded under the attitude verb *san* ‘to think’ as in (29), where *indexical shift* is possible. Again, if the indexical 1st person and the evidentiality can shift independently, we predict that (29) should have four different logical interpretations. This prediction is not borne out. The subscripts next to ‘apparently’ indicate who lacks direct evidence for the purchase.

- (29) If *Dursun* is speaking:
 Burak (ben) yanlışlıkla ev al-mış-ım san-ıyor.
 Burak 1SG wrong-NMLZ-INS house buy-evid-1SG think-prog
- ‘Burak thinks that **speaker apparently**_{spkr} bought a house by mistake.’
 - ‘Burak thinks that **Burak apparently**_{Burak} bought a house by mistake.’
 - *‘Burak thinks that **speaker apparently**_{Burak} bought a house by mistake.’
 - *‘Burak thinks that **Burak apparently**_{spkr} bought a house by mistake.’

The first available interpretation of (29) is where neither the 1st person indexical nor the evidentiality is shifted (29-a). In this reading, the indexical refers to the speaker *Dursun*, and the evidentiality is also evaluated based on *Dursun*’s (speaker) perspective. This leads to a reading like ‘Dursun made a purchase, and Dursun lacked evidence for the purchase’. (29) could be uttered naturally in this reading to report that *Burak* believes in the situation reported in Context 1. The second available interpretation of (29) is where both the 1st person indexical and the evidentiality are shifted (29-b). So, in this reading, the 1st person indexical refers to the attitude holder *Burak*, and the evidentiality is also evaluated based on *Burak*’s perspective. This leads to a reading like ‘Burak made a purchase, and Burak lacked evidence for the purchase’. (29) could naturally be uttered in this reading to describe the situation described in Context 2, where the purchase is made by Burak, and Burak lacks direct evidence for the purchase (but not the speaker).

- (30) Context 2: *Burak sleeps in the living room, where Dursun is watching TV. Burak has a nightmare, in which he presses some buttons on his computer and ends up buying a house by mistake. When he wakes up, he does not notice that it was a dream and he believes that he really bought a house by mistake. Burak goes: ‘OMG, I bought a house by mistake!’. Dursun is aware that Burak just had a dream, and knows that he actually did not buy a house.*

Similar to what is observed with the motion verb *gel* ‘to come’ in the previous subsection, we see here that both available interpretations require that the indexical *ben* and the PSI *gel* get interpreted relative to the same context: either the context of utterance (29-a), or the reported context (29-b). Readings where only one of these two elements shifts are unavailable (29-c)-(29-d), even in contexts where those readings would be expected to be felicitous. One such context would be Context 3, where *Burak* has direct evidence for his purchase, but *Dursun* lacks direct evidence.

- (31) Context 3: *Burak, a very poor friend of Dursun, claims that he bought a house. Burak claims that he met the seller and received the keys from them (Burak has direct evidence for the purchase). However, Dursun is skeptical about this and does not fully believe that the story is true (Dursun lacks direct evidence).*

Crucially, a sentence like (29) cannot be accurately uttered by *Dursun* to describe the situation in Context 3. This is mainly due to the fact that this interpretation would force the 1st person indexical to refer to *Burak* (shifted), while using an evidential marker from the perspective of the speaker *Dursun* (non-shifted). Unavailability of (29) to describe a situation like Context 3 shows that the 1st person indexical and the evidentiality cannot be interpreted inside different contexts, even in situations that would allow/require such readings.

In sum, echoing what we saw with the motion verb *gel* ‘to come’, *perspective shift* of evidential markers is correlated with *indexical shift* in Turkish finite embedded clauses. When *indexical shift* doesn’t take place, evidential markers cannot be shifted. When indexical shift takes place, evidential markers must also shift. In other words, the evidential markers and the 1st person indexicals must ‘shift together’. This, again, follows naturally under the analysis developed in this paper. If evidentiality is evaluated relative to the world knowledge (amount of evidence) of the *author*, it is expected to shift whenever indexical shift (of the 1st person) replaces the the *author* with the attitude holder, and is expected to stay non-shifted in the absence of indexical shift.

Data introduced here shows that the motion verb *gel* ‘to come’ and evidential markers both show shift together patterns with the 1st person indexical and form the first sub-type of PSIs (Type 1), which I call *Indexical PSIs*. As explained in more detail below, I suggest that Indexical PSIs have semantic definitions that refer to the speech-act coordinates of the context (e.g. *author*), as in (7), which makes them targeted by the indexical shifting operator. If semantic definitions along the lines of those are adopted, the behavior of indexical PSIs can be accounted by with the standard, OP-based analyses of indexical shift (e.g. Anand and Nevins, 2004; Deal, 2020).

3.2 Non-indexical PSIs: PSIs that do not shift together with indexicals

Our investigation of various types of PSIs reveals that not all PSIs show indexical-like properties. In this section, we provide data with epistemic modals, predicates of personal taste, relative socio-cultural expressions, and epithets in indexical shifting contexts, and show that they behave differently from indexical PSIs. Though these PSIs are not typically considered to form a uniform group (i.e. they have different properties within Turkish and cross-linguistically), the relevant observation for the purposes of this study is that they are all insensitive to indexical shift, which implies that they are not interpreted relative to the indexical *author* parameter of the context.

Relative socio-cultural expressions like *local*, *foreigner*, *enemy*, and *fellow citizen* are interpreted based on the socio-cultural status of a perspective holder (Mitchell, 1986; Partee, 1989; Oshima, 2006; Bylina et al., 2014), so a word like *foreigner* means something like ‘not from the same country with the perspective holder’. The perspective holder of the relative socio-cultural expressions (henceforth, relative expressions) in root clauses is typically the speaker by default, but they can undergo perspective shift and be interpreted from the perspective of another person.

Sentences like (32) allow us to test any possible interactions between relative expressions and indexical shift. (32) has an indexical pronoun like 1st person *ben* and an SCE like *yerel* ‘local’ in the embedded clause. Considering that both of these items can optionally shift, a sentence like (32) has four different logically possible readings. If indexical shift and the perspective shift of relative expressions are independent from each other, we predict that the sentence should be ambiguous in four different readings. However, if indexical shift and perspective shift interact and they must (or cannot) take place together, the sentence should not be ambiguous in more than two different readings, as we observe with indexical PSIs.

- (32) If *Dursun_i* is speaking:
 Ken (**ben**) **yere**l bir gazete-ye röportaj ver-di-m san-ıyor.
 Ken 1SG local a journal-DAT interview give-PST-1SG think-PROG
 ‘Ken thinks that I/he gave an interview to a local journal.’
 a. → Ken thinks *Dursun* interviewed a Turkish journal (local to *Dursun*).
 b. → Ken thinks *Ken* interviewed a Japanese journal (local to *Ken*).
 c. → Ken thinks *Dursun* interviewed a Japanese journal (local to *Ken*).
 d. → Ken thinks *Ken* interviewed a Turkish journal (local to *Dursun*).

We see that (32) can be interpreted in four different ways, depending on whether indexical shift and perspective shift independently occurred. More precisely, we see that both items can stay non-shifted as in (32-a) (interpreted from the perspective of the speaker), they can both shift at the same time as in (32-b) (interpreted from the perspective of the attitude holder), or one of them may shift while the other does not, as in (32-c) and (32-d). This shows that indexical shift and perspective shift of the relative expression *yere*l ‘local’ are independent from each other, and one can take place even if the other does not. This implies that the interpretation of some PSIs like relative expressions is not sensitive to the indexical coordinates of the context, and that PSIs should be categorized into two as *indexical PSIs* and *non-indexical PSIs*.

Further evidence for the distinction of indexical shift and perspective shift of PSIs comes from examples with epistemic modals. The semantics of epistemic modals has been largely studied in the literature. Though there are differences in the semantic details of how epistemic modals are interpreted, it has been reported that epistemic modals can have shifted or non-shifted readings inside attitude reports (e.g. Egan et al., 2005; Stephenson, 2007; Yanovich, 2014; Roberts, 2023; Rudin, 2024). Though the current study does not aim to provide any detailed analysis of how epistemic modals are interpreted in Turkish, our preliminary investigation suggests that by epistemic modals in Turkish are by default interpreted from the perspective of the speaker in matrix clauses and from the perspective of the attitude holder (at least in stand alone sentences without rich contextual information). This is demonstrated for Turkish in examples (33-a) and (33-b).

- (33) a. Salı gün-ü yağmur yağ-**abil**-ir.
 Tuesday day-3SG.POSS rain fall-MOD-AOR
 ‘It may rain on Tuesday.’
 → It is compatible with *speaker*’s knowledge that it is possible that it rains on Tuesday.
 b. Burak Salı gün-ü yağmur yağ-**abil**-ir san-ıyor.
 Burak Tuesday day-3SG.POSS rain fall-MOD-AOR think-PROG
 ‘Burak thinks that it may rain on Tuesday.’
 → It is compatible with *Burak*’s knowledge that it is possible that it rains on Tuesday.

While the modal in (33-a) is interpreted relative to the knowledge of the speaker, the modal in (33-b) undergoes perspective shift and gets interpreted relative to the knowledge of the attitude holder *Burak*. Native speakers report that non-shifted readings are not available (at least easily) in (33-b), suggesting that the default interpretation of epistemic modals shift in Turkish attitude reports is from the perspective of the attitude holder.

In light of this, sentences like (34), with a 1st person indexical and an epistemic modal in the embedded clause, show that the perspective shift of the modals and indexical shift are independent from each other.

- (34) If *Dursun* is speaking:
 Burak (**ben**) hasta ol-**abil**-ir-im san-ıyor.
 Burak 1SG sick be-MOD-AOR-1SG think-PROG
- a. ‘Burak thinks that I may be sick.’
 → It is compatible with *Burak*’s knowledge that it is possible that *Dursun* is sick.
- b. ‘Burak thinks that he may be sick.’
 → It is compatible with *Burak*’s knowledge that it is possible that *Burak* is sick.

As in (33-b), we see that the modal in (34) shifts by default and is interpreted under the perspective of the attitude holder *Burak*. However, the indexical can stay non-shifted and refer to the speaker *Dursun* (34-a), or it can shift and refer to the attitude holder *Burak* (34-b). In other words, the indexical can stay non-shifted when the epistemic modal shifts, suggesting that perspective shift of modals and indexical shift are independent mechanisms. This contrasts recent work by Roberts (2023), who suggests that epistemic modals could be indexical.

A similar pattern is observed with predicates of personal taste (PPTs) as in (35), where a (dropped) 1st person possessive pronoun and the PPT *lezzetli* ‘tasty’ appear inside the same embedded clause.

- (35) If *Dursun_i* is speaking:
 Burak **pro**.1sg anne-m-in suşi-si çok **lezzetli** san-ıyor.
 Burak mother-1SG.POSS-GEN sushi-3SG.POSS very tasty think-PROG
- a. ‘Burak thinks that **my** mother’s sushi is very tasty.’
 → Sushi is judged to be very tasty by *Burak* (attitude holder).
- b. ‘Burak thinks that **his** mother’s sushi is very tasty.’
 → Sushi is judged to be very tasty by *Burak* (attitude holder).

Though the PPT shifts and is interpreted from the perspective of the attitude holder *Burak* by default⁷, the 1st person possessor can stay non-shifted (35-a), referring to the speaker, or it can shift and refer to the attitude holder *Burak* (35-b). So, again, the 1st person indexical can stay non-shifted even when the PPT undergoes perspective shift (35-a), suggesting that the two phenomena are independent from each other.

Another piece of evidence for the separation of indexical shift and perspective shift of some PSIs comes from epithets. Epithets (e.g. the idiot, the bastard) express an evaluation or judgment towards their referents, and must be interpreted under the perspective of a judge/evaluator (e.g. Kratzer, 1999; Speas and Tenny, 2003; Potts, 2004; Amaral et al., 2007; Harris and Potts, 2009). Our investigation suggests that epithets in Turkish cannot shift inside attitude reports (at least, cannot shift easily⁸), meaning that a speaker cannot use an epithet unless they share the judgment/evaluation towards its referent⁹. For example, the use of the epithet *şerefsiz* ‘scumbag’ in (36) indicates that the speaker (but not necessarily the attitude holder) thinks *Ali* is a scumbag. However, this does not seem to interact with indexical shift, as the 1st person indexical in the sentence can stay non-shifted (36-a) or shift (36-b).

⁷ As discussed in Stephenson (2007) and subsequent work, predicates of personal taste (PPTs) in attitude reports *can* be interpreted from the perspective of a judge other than the attitude holder, as in sentences like ‘My wife thinks that this cat food is tasty’, where *tasty* is naturally interpreted as ‘tasty for the cat’. Parallel examples exist in Turkish as well. The relevant fact here is that PPTs are generally interpreted from the perspective of the attitude holder in standalone sentences without specific semantics/pragmatics.

⁸ Previous theoretical and experimental work suggests that epithets are by default interpreted as expressing the evaluations of the speaker, but the interpretation can shift in exceptional scenarios that require very rich contextual set ups (e.g. Harris and Potts, 2009; Patel-Grosz, 2012; Kaiser, 2015).

⁹ Some speakers note that they might take sentences like (36) to possibly express the judgment from the attitude holder’s perspective, if the speaker does not have any particular judgment towards *Ali* (e.g. they do not know *Ali* or they barely know anything about him). However, if they know *Ali* and disagree with the judgment that he is a scumbag, they say they cannot utter (36). For those speakers, including the author of the current study, it might be the case that they allow perspective shift of an epithet in specific contexts (e.g. when they do not have a judgment of the referent of the epithet), in line with the behavior reported for English (e.g. Harris and Potts, 2009; Patel-Grosz, 2012; Kaiser, 2015). Most speakers, however, provided consistent judgments indicating that they do not allow perspective shift of an epithet.

- (36) If *Dursun* is speaking:
 Burak (**ben**) Ali şerefsiz-i-ni gör-dü-m san-ıyor.
 Burak 1SG Ali scumbag-3.POSS-ACC see-PST-1SG think-PROG
- a. ‘Burak thinks that **I** saw that scumbag Ali.’
 → *Dursun* (speaker) thinks Ali is a scumbag (non-shifted).
- b. ‘Burak_i thinks that **he_i** saw that scumbag Ali.’
 → *Dursun* (speaker) thinks Ali is a scumbag (non-shifted).

The fact that the indexical can shift while the epithet does not, as in (36-b), suggests that the perspective shift of epithets and indexical shift are two independent mechanisms.

In overall, we see that the shifting patterns of some PSIs like epistemic modals, predicates of personal taste, relative socio-cultural expressions, and epithets are not affected by the (non-)occurrence of indexical shift. As we have seen through examples (32)-(36), the interpretations of these PSIs do not change when indexical shift occurs. This suggests that these PSIs do not have indexical-like properties, unlike *indexical PSIs*, thus we categorize them as *non-indexical PSIs*.

In the next subsection I provide a summary of the patterns observed with different types of PSIs and develop an analysis to account for those patterns.

3.3 Interim summary Analysis

In summary, in the preceding sections I presented new empirical data demonstrating that perspective sensitive items (PSIs) do not exhibit a uniform behavior regarding their sensitivity to indexical shift. Data illustrated that the motion verb *gel* ‘to come’ (24) and evidential markers (29) *must* be shifted whenever they appear in a domain where indexical shift takes place, and they *cannot* be shifted if indexical shift does not take place. This suggests that the interpretation of the motion verb *gel* ‘to come’ and evidential markers are sensitive to indexical shift, and that they must *shift together* with the 1st person indexicals. Since the interpretations of these PSIs are tied together with indexicals, we call them *indexical PSIs*. Other PSIs, which I call *non-indexical PSIs*, on the other hand, do not seem to be affected by whether indexical shift takes place or not. For example, relative socio-cultural expressions (e.g. *local*) can freely shift independently from indexical shift (32), epistemic modals (e.g. *may/might*) shift by default even in cases where indexical shift does not take place (34), and epithets (e.g. *idiot*) shift even when they appear in an indexical shift environment (36).

The existence of *indexical PSIs* is surprising, considering that PSIs are traditionally treated as a different class of items than indexicals, and have been proposed to be interpreted relative to non-indexical coordinates like *judge* (Lasersohn, 2005; Potts, 2007; Amaral et al., 2007; Bylinina et al., 2014; Kaiser, 2022a, a.o.). For example, Lasersohn (2005) suggests that predicates of personal taste (PPTs) are interpreted relative to the *judge* coordinate of the context, which represents the individual whose perspective the PPT is ‘judged’ from. Stephenson (2007) extends this analysis with modifications to epistemic modals (but see (Kratzer, 1977; Yalcin, 2007) for alternative accounts), and Potts (2007) extends it to epithets. Though the specific details of these accounts vary, the important matter relevant to the current study is that those accounts all suggest that these PSIs are not interpreted relative to the *author* coordinate of the context. Thus, according to this view, the interpretations of indexicals and PSIs should be independent from each other, and shift in one domain (e.g. indexical shift) should not be able to force or block shift in the other domain (e.g. perspective shift). Though an approach like this is compatible with the patterns observed with some PSIs (e.g. PPTs, epithets), the patterns observed in Turkish with *indexical PSIs* suggest that at least some types of PSIs are interpreted in a similar way to indexicals, that is, they are sensitive to the indexical coordinates of the context. Based on this, I suggest that the motion verb *gel* ‘to come’ and the evidential markers in Turkish are interpreted relative to the *author* coordinate of the context, rather than *judge*, and thus they must have shifted readings when they appear in an indexical shifting environment. This is in line with the predictions of some of previous work, which suggested that PSIs might possibly have indexical properties and at least some types of perspective shift could be analyzed as indexical shift (e.g. Amaral et al., 2007; Percus, 2011). However, this idea was abandoned mostly based on English data like (37), where PSIs can have shifted interpretations even though indexical elements cannot shift.

- (37) My father screamed that he would never allow **me** to marry that **bastard** Webster.
(from Kratzer, 1999, ex (14))

The famous example in (37), originally from Kratzer (1999), discussed in Amaral et al. (2007), shows that the 1st person indexical *me* is interpreted as referring to the speaker (non-shifted), while the epithet *bastard* is shifted and interpreted from the perspective of the attitude holder ‘father’ (e.g. the speaker does not think Webster is a bastard). Considering that English does not allow indexical shift (as evident by the non-shifted *me*), (37) implies that the perspective shift of PSIs (e.g. *bastard*) is not due to indexical shift and it must be independent from indexical shift.

In a highly relevant study, Sudo (2018) takes a different approach and suggests that English allows indexical shift in presuppositional context, and some PSIs (e.g. *come*) can have shifted interpretations as a result of indexical shift. Sudo (2018) reports data in (38), demonstrating that English *come* can have shifted interpretations in attitude reports.

- (38) Natasha said that Chomsky is *coming* to New Jersey next week.
a. → Speaker is in New Jersey (but Natasha need not be).
b. → Natasha is in New Jersey (but speaker need not be).
(from Sudo, 2018, ex. (40))

In (38), the verb *come* can either imply that the speaker is in New Jersey (non-shifted), or imply that the attitude holder *Natasha* is in New Jersey (shifted). Sudo (2018) proposes that English *come* has an ‘indexical presupposition’ that needs to be satisfied. In simple terms, Sudo argues that *come* presupposes that there is a moving event towards the location of the *author* (speaker), and that the verb *come* can only be used to describe a moving event when this presupposition is satisfied. Whenever this presupposition is not satisfied (e.g. movement is not towards the *author*), the alternative motion verb *go* should be used instead.

Under Sudo’s account, (38-b) is an instance of (presuppositional) indexical shift, where the *author* of the context for presupposition of *come* is *Natasha*. Since English does not allow indexical shift, Sudo suggests that presuppositions refer to a different set of context parameters (let’s call it *context*₂), which can be shifted even in languages that does not allow regular indexical shift. According to this, the *author* of (38-b) is still the original speaker of the sentence (author of *context*), but the presupposition of *come* should be satisfied in a shifted context (*context*₂), where *author* is *Natasha*. This mechanism successfully accounts for the shifted behavior of *come*, while not requiring/allowing indexical pronominals (e.g. *I*, *me*, *my*) to shift.

This extra mechanism that Sudo develops is compatible with (but not required for) the Turkish data introduced here. In Turkish data provided here, we observe that what described as presuppositional indexical shift (e.g. *gel* ‘to come’) must occur together with regular indexical shift (e.g. 1st person). Moreover, there is no reliable diagnostic to test if the shift observed with *gel* ‘to come’ is presuppositional or regular indexical shift. Considering these, I simply assume that standard (not presuppositional) indexical shift is responsible for the shifty behavior of *gel* ‘to come’ in Turkish. The technical details of the semantics of motion verbs is out of the scope of this paper, so I do not provide a detailed semantic discussion. In broad terms, any semantic denotation that ties the interpretation of the motion verb *gel* ‘to come’ to the *author* parameter of the context (as Sudo does) could account for the shift together facts of *gel* and 1st person indexicals in Turkish. I leave the semantic details for future work.

A similar mechanism should be at play for evidential markers, where evidentiality is interpreted with reference to an indexical context parameter like *author*. There is a large literature which suggests that evidentiality is evaluated/interpreted based on the world knowledge of the speaker a sentence (Izvorski, 1997; Speas and Tenny, 2003; Sauerland and Schenner, 2007; Stephenson, 2007, a.o.). For example, Izvorski (1997) suggests that the use of an indirect evidential marker asserts that the reported proposition *p* is compatible with *the speaker*’s knowledge, and it presupposes that *the speaker* has indirect evidence for *p*. This is similar to Sudo’s proposal for *come*, discussed above, in that the interpretation of the evidential marker refers to the context coordinate *author*, which is sensitive to indexical shift. Though there are different views on the details of the semantics of evidentiality, any approach that ties the interpretation of evidential markers to the context coordinate *author* should successfully account for the shift together facts of evidentials and 1st person indexicals in Turkish.

Leaving the details of their semantics for future work, I propose that the motion verb *gel* ‘to come’ and the evidential markers like ‘-*mİş*’ in Turkish are sensitive to the indexical coordinate *author*. I suggest that the semantics of *gel* and the evidential marker ‘-*mİş*’ is along the lines of (39-a) and (39-b)¹⁰.

- (39) a. *gel* = an event of movement towards the *author*
 b. ‘-*mİş*’ = the *author* lacks direct evidence for reported proposition

If semantic descriptions along the lines of (39) are adopted, the shifting (and shift together) facts of indexical PSIs can be straightforwardly captured under a standard context-overwriting accounts of indexical shift (e.g. Anand and Nevins, 2004; Deal, 2020). I follow previous work in Turkish (Özyıldız et al., 2018; Akkuş, 2019; Oğuz et al., 2020) and propose that a context overwriting operator OP_{AUTH} can merge in the left periphery of the finite embedded clauses in Turkish and replace the *author* parameter of the context with the attitude holder, schematized in (40).

- (40) Speaker = *Dursun_i*
 a. [TP Burak_k [CP/TP 1SG_i ... *come_i/evidentials_i*] think] non-shifted
 b. [TP Burak_k [_{scp}/TP OP_{AUTH} [1SG_k ... *come_k/evidentials_k*]] think] shifted

The merge of this OP_{AUTH} changes the context under which indexicals (1st person) and indexical PSIs (*gel* ‘to come’ or evidential markers) are interpreted, replacing the value of the *author* coordinate with the attitude holder (40-b). In the absence of this OP_{AUTH} , all of these items are interpreted in the actual speech context, where the speaker is the *author* (40-a). Assuming that the semantics of *gel* and ‘-*mİş*’ are roughly as in (39-a) and (39-b), the shift together facts observed in sentences like (24) and (29), repeated here as (41) and (42), are expected since the merge of OP_{AUTH} changes the context that the indexicals and indexical PSIs are interpreted in.

- (41) If *Dursun* is speaking in *Los Angeles*
 Ev arkadaş-ım [CP (OP_{AUTH}) (**ben**) Boston-a *gel-ey-im*] ist-iyor.
 House mate-1SG 1SG Boston-DAT come-OPT-1SG want-PROG
 a. ‘My roommate wants that I come to Boston_{#SPKR/ADDR}.’ (neither shifts)
 b. #‘My roommate wants that he comes to Boston_{ROOMMATE}.’ (both shift)

In (41), the indexical shifting operator OP_{AUTH} can merge in the left periphery of the embedded clause and shift the *author* parameter of the context in which the sentence is interpreted. As a result, all indexical elements that are sensitive to the *author* parameter, 1st person *ben* and *gel* ‘to come’ in this case, will be interpreted in a context, where the roommate is the *author*, leading to the interpretation in (41-b). When this OP_{AUTH} is absent, the sentence is interpreted in the actual speech context, where both 1st person *ben* and *gel* ‘to come’ are interpreted as referring to the speaker *Dursun* as the *author*, and the addressee is available for listener-anchoring (41-a). Crucially, since both 1st person *ben* and *gel* ‘to come’ depend on the *author* parameter of the context, they can either shift together or they cannot shift at all.

- (42) If *Dursun* is speaking:
 Burak [CP (OP_{AUTH}) (**ben**) yanlışlıkla ev al-mış-ım] san-ıyor.
 Burak 1SG wrong-NMLZ-INS house buy-evid-1SG think-prog
 a. ‘Burak thinks that Dursun apparently bought a house by mistake!’ → Dursun lacks evidence
 b. ‘Burak thinks that Burak apparently bought a house by mistake!’ → Burak lacks evidence

Similar to (41), a context-overwriting OP_{AUTH} can optionally merge in the embedded clause of (42), and replace the *author* parameter of the context in which the sentence is interpreted with the attitude holder *Burak*. This forces the 1st person indexical and the evidential marker ‘-*mİş*’ to be interpreted in a context where *Burak* is the *author*, and leads to the shifted interpretation in (42-b). In the absence of the OP_{AUTH} , the sentence is interpreted in the actual speech context, where the actual speaker *Dursun* is the *author*,

¹⁰ These are descriptive explanations of a semantic analysis that would be compatible with the Turkish data introduced in this paper, and are not intended as formal proposals. I leave the semantic/formal details of these items for future work.

leading to the interpretation in (42-a). Again, since both 1st person *ben* and ‘-*mİş*’ depend on the same context coordinate, they can either shift together, or not shift at all.

Importantly, the analysis developed here suggests that the shifted readings of indexical PSIs (e.g. the motion verb *gel* ‘to come’ and evidential markers) are due to the occurrence of indexical shift, rather than perspective shift (of the *judge*), which allows shifted readings for other types of PSIs. As a result, this approach predicts that indexical PSIs cannot have shifted readings in environments where indexical shift is not available. This prediction is borne out, and can be demonstrated by examples with nominalized embedded clauses. It has been well documented that nominalized embedded clauses like (43) in Turkish do not allow indexical shift (e.g. Şener, 2011; Özyıldız, 2012). In (43), the 1st person embedded subject must be interpreted as referring to the speaker, and cannot shift and refer to the matrix subject *Ali*.

- (43) Ali (**ben-im**) Boston-a git-tiğ-**im-i** san-ıyor.
 Ali 1SG-GEN Boston-DAT go-NMLZ-1SG-ACC think-PROG
 ‘Ali thinks that {**I/*he**} went to Boston.’

Crucially, even though indexical shift is not allowed in nominalized clauses, (non-indexical) PSIs can still undergo perspective shift. This is demonstrated in (44), with the relative socio-cultural expression *yerel* ‘local’. Assuming that Dursun is Turkish, Ken is Japanese, and John is American, we see that the relative expression *yerel* ‘local’ can be anchored to the speaker (Turkish), to the embedded subject (Japanese), or to the matrix subject (American).

- (44) If *Dursun* is speaking:
 John Ken-in **yerel** bir gazete-ye röportaj ver-diğ-i-ni san-ıyor.
 John Ken-GEN local a journal-DAT interview give-NMLZ-3SG-ACC think-PROG
 ‘John thinks that Ken interviewed a local_{TURKISH/JAPANESE/AMERICAN} journal.’

Examples like (43) and (44) show that indexical shift is not allowed in nominalized embedded clauses, while perspective shift (of PSIs) is allowed. So, if the motion verb *gel* ‘to come’ behaves like other PSIs, we would expect it to be able to shift inside nominalized embedded clauses. However, if *gel* behaves like indexicals (and its shift is due to this property, as proposed here), we predict it to be unable to shift inside nominalized clauses. This prediction is borne out, as demonstrated in (45).

- (45) a. Ali (**ben-im**) Boston-a **gel-diğ-im-i** san-ıyor.
 Ali 1SG-GEN Boston-DAT go-NMLZ-1SG-ACC think-PROG
 ‘Ali thinks that {**I/*he**} came to Boston.’ → ‘Spkr (or Addr) is in Boston’
 b. Ali Ayşe-nin Boston-a **gel-diğ-in-i** san-ıyor.
 Ali Ayşe-GEN Boston-DAT go-NMLZ-3SG-ACC think-PROG
 ‘Ali thinks that Ayşe came to Boston.’ → ‘Spkr (or Addr) is in Boston’

Sentences in (45) have the motion verb *gel* ‘to come’ inside nominalized embedded clauses. Both sentences in (45) imply that the speaker (or addressee, via listener-anchoring) should be in Boston, and they cannot be uttered in a context where neither the speaker nor the addressee is in Boston. Importantly, these sentences do not have any implications regarding the location of the matrix subject *Ali*, suggesting that *gel* cannot shift in this environment. This is different from finite embedded clauses (41), where indexical shift is available. Here, we see that the unavailability of indexical shift blocks shifted readings of *gel* ‘to come’, with (45-a) or without (45-b) an indexical element in the same domain. This provides further evidence for the analysis developed here, suggesting that *gel* ‘to come’ can shift due to its indexical properties, only when it appears in an indexical shifting environment.¹¹

In the next section, I provide novel data regarding Accusative Subject Clauses (ASCs) in Turkish and suggest that indexical shift is observed in the embedded clauses in ASCs as well, and I extend the analysis developed here to include ASCs.

¹¹ A similar test cannot be applied to evidential markers, since the evidential marker ‘-*mİş*’ cannot attach nominalized clauses (e.g. Korotkova, 2019).

4 Indexical Shift in Accusative Subject Clauses

Turkish allows finite embedded clauses where the agent/subject of the embedded verb is marked with an Accusative marker, as the 1st person indexical *ben* in (46). I refer to such sentences as *Accusative Subject Clauses* (ASCs). In this section, I provide data with ASCs and extend my analysis to cover the facts observed with ASCs. Based on novel data I introduce in this section I conclude that the optional agreement marker on the embedded verb of ASCs indicates whether indexical shift takes place in the embedded clause or not. More precisely, I suggest that the lack of a full, matching agreement marker between the accusative-marked subject and the embedded verb indicates the merge of the indexical shifting OP_{AUTH} into the embedded clause, forcing all indexical items (including the indexical PSIs) within its scope to shift.

- (46) Burak **ben-i** Ankara-ya git-ti-(m) san-ıyor.
 Burak 1SG-ACC Ankara-DAT go-PST-1SG think-PROG
 ‘Burak thinks that I/*he went to Ankara.’

Three key aspects of ASCs are particularly relevant for the current study.

First of all, the previous literature reports that the embedded verb agreement in ASCs is optional, without any discussion of the semantic differences correlated with the agreement (Şener, 2008; Özgen and Aydın, 2016; Başoğlu, 2021). According to these accounts, the agreement on the embedded verb does not affect the interpretation of the sentence, and a full agreement is projected when the embedded T/C heads have the full set of phi-features but agreement is not projected when embedded T/C heads are defective (lacking phi-features to trigger agreement). Using novel data, I show below that the optionality of embedded verb agreement is not totally optional, and it is correlated with indexical shift.

Second, as firstly noted by Şener and Şener (2011), the accusative-marked 1st person indexical *ben* in (46) can only be interpreted as referring to the speaker, not allowing shifted readings. A parallel structure exists in Uyghur, where the accusative-marked subject cannot undergo indexical shift (Shklovsky and Sudo, 2014; Major, 2022). Şener and Şener (2011) and Shklovsky and Sudo (2014) suggest that in Turkish and Uyghur, respectively, it is not clear if indexical shifting operator (OP) could merge in the embedded clauses of ASCs, and they argue that even if the OP merged in the structure the accusative-marked subject would be outside the scope of it and thus it would remain non-shifted. In a more recent study, Major (2022) argues that Uyghur ASCs lack the syntactic structure to host the OP (e.g. they are syntactically reduced) and thus the OP cannot merge into ASCs. However, it is still unknown whether the OP can be present in the embedded clauses of ASCs in Turkish. The unavailability of shifted readings of the accusative-marked subject can be due to the fact that it is outside of the scope of the OP (as implied by Şener and Şener), or simply due to the fact that the OP does not merge in the structure with the ASCs (as proposed for Uyghur by Major). Based on new empirical facts, I suggest that the OP indeed merges to the left periphery of the embedded CP in Turkish, but the accusative-marked subject is located in (merges directly or moves into) a higher position than the OP and thus is not affected by its merge.

Third, ASCs are traditionally analyzed as *Subject-to-Object Raising* (SOR) or *Exceptional Case Marking* (ECM) structures, schematized in (47-a) and (47-b), respectively. According to the SOR approach, the embedded subject raises higher into the matrix object position to check case feature (Zidani-Eroğlu, 1997; Aygen, 2002; Başoğlu, 2021). The ECM approach, on the other hand, suggests that the embedded subject element raises into the left periphery of the embedded CP, and checks accusative case non-locally from inside the embedded CP (Şener, 2008; Özgen and Aydın, 2016; Predolac, 2017). For the Uyghur counterparts of ASCs, Major (2024) suggests that a proleptic structure is available, where the so-called accusative subject (DP_{ACC}) merges as the matrix object and is in coreference with another DP (usually silent *pro*) in the embedded clause. Considering the similarities between Uyghur and Turkish, I entertain the proleptic analysis as a third possibility. I schematize each of these possible analyses below in (47) in light of the proposal developed here.

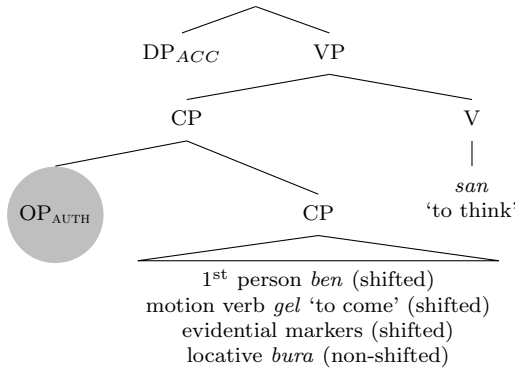
- (47) a. [TP Burak [vP/VP DP_{ACC_i} [CP/TP (OP_{AUTH}) [t_i Ankara go]] think]] SOR
 b. [TP Burak [vP/VP [CP DP_{ACC_i} [CP/TP (OP_{AUTH}) [t_i Ankara go]] think]]] ECM
 c. [TP Burak [vP/VP DP_{ACC_i} [CP (OP_{AUTH}) [TP DP_i Ankara go]] think]] Prolepsis

Though the syntactic status of ASCs is not in the scope of the current paper, I entertain those three possible analyses and show how the indexical shifting observations should be accounted for in line with each one. As discussed in the following subsections in detail, I argue that the indexical shifting OP_{AUTH} must merge the embedded clause periphery in ASCs. So, if SOR or ECM analyses of ASCs are adopted, this would require that the OP_{AUTH} merges to the periphery of the embedded CP, and later the embedded subject raises higher than OP_{AUTH} to satisfy case requirements and escapes the scope of the OP_{AUTH} (thus stays non-shifted). However, under the proleptic analysis, the accusative subject merges higher than the OP_{AUTH} , and thus is not affected by its merge. Major (2022, 2024) concludes that indexical shift is not available in ECM (or SOR) structures because they lack the syntactic structure (non-defective C/T layers) to host the OP. If we take this to be true in Turkish as well, this would favor the proleptic analysis of ASCs over the ECM and SOR ones.

Regardless of which syntactic analysis is adopted, the merge of the OP_{AUTH} predicts all (1st person) indexical elements inside the embedded clause to have shifted interpretations, while a 1st person accusative subject/agent stays unshifted as a result of its syntactic status. In this section, I use *indexical PSIs* introduced in Section 3 (e.g. motion verbs, evidential markers) and possessive pronouns and show that this prediction is borne out. Precisely, I show that these items have shifted interpretations when the embedded verb lacks full, matching agreement with the accusative-marked subject/agent. Though the current paper stays neutral about which syntactic analysis should be adopted, it is compatible with all three analyses discussed above. The proleptic analysis is different from the other two in that it requires full CP structures that bear agreement morphology, while ECM/SOR analyses predict agreement to be optional. This means that ECM/SOR analyses allow defective C/T layers that do not license agreement at all, while the proleptic analysis does not allow this. Considering that the 3rd person agreement marker is morphologically null in Turkish, the apparent lack of agreement on embedded verb can be considered as full agreement with a 3rd person agreement. Under such an analysis, the accusative-marked subject would be in coreference with a 3rd person pronominal (which can be silent) in the embedded clause, and this 3rd person element would control embedded verb agreement.

I schematize my proposed structure in (48) for the embedding verb *san* ‘to think’ and I extend it to cover the shifty behavior with the other embedding verb *de* ‘to say’ at the end of this section.

(48) *Shifty OP embedded under san* ‘to think’



In the rest of the paper, I stick to the traditional accounts of ASCs in Turkish (e.g. Zidani-Eroğlu, 1997; Şener, 2008; Predolac, 2017) and use glossing following the ECM/SOR type of analyses. I avoid providing extra glossing in line with the proleptic analysis for space reasons.

4.1 Motion verb *gel* ‘to come’

This subsection provides novel data with the motion verb *gel* ‘to come’ inside ASCs and shows that there is a direct correlation between embedded verb agreement choice and the interpretation of the sentence. When there is matching/full 1st person agreement on the embedded verb, *gel* is interpreted in a non-shifted way. When the default (null) agreement is used, *gel* is shifted. We conclude from this that the indexical shifting OP merges in ASCs without matching/full agreement and shifts the interpretation of *gel* ‘to come’.

The previous sections showed that the motion verb *gel* ‘to come’ behaves like indexicals and has shifted interpretations when it appears under the scope of the indexical shifting OP_{AUTH} . Based on previous data and observations, I suggested that *gel* should have a semantic definition roughly like (39-a), repeated here as (49).

- (49) *gel* = an event of movement towards the *author*

Based on this definition, we predict that *gel* describes a movement towards the speaker or the addressee (via listener anchoring) in environments where indexical shift does not take place, and it describes a movement towards the attitude holder in environments where indexical shift takes place. This is due to the fact that indexical shift replaces the *author* parameter of the context with the attitude holder, and the attitude holder is the *author* of this new context, relative to which *gel* will be interpreted. In light of this, consider the contrast in (50), where the choice of the embedded verb agreement seems to affect how the motion verb *gel* is interpreted.

- (50) If *Dursun* is speaking:

- a. Burak **ben-i** Ankara-ya gel-di-m san-ıyor.
 Burak 1SG-ACC A.-DAT come-PST-1SG think-PROG
 ‘Burak thinks that I came to Ankara.’ → ‘Dursun (or addressee) is in Ankara’
- b. Burak **ben-i** Ankara-ya gel-di-Ø san-ıyor.
 Burak 1SG-ACC A.-DAT come-PST think-PROG
 ‘Burak thinks that I went to Ankara.’ → ‘Burak is in Ankara’

Both sentences in (50) describe a belief of *Burak* that the speaker *Dursun* has made a movement towards Ankara, but they differ minimally in that (50-a) bears embedded verb agreement that matches the accusative subject DP (1st person) while (50-b) lacks the matching agreement marker. However, the sentences differ in a crucial way in their interpretations. While (50-a) implies that the speaker *Dursun* (or his addressee, via listener-anchoring) is in Ankara, (50-b) implies that the attitude holder *Burak* is in Ankara. In other words, in (50-a) the motion verb describes a movement towards the speaker, who is the default perspective holder, but in (50-b) it describes a movement towards the attitude holder *Burak*. This suggests that while the *author* of the context is the speaker in (50-a), as the default, the *author* is the attitude holder *Burak* in (50-b), as a result of indexical shift.

Based on this contrast, it appears that indexical shift occurs in ASCs without embedded verb agreement that matches the phi-features of the accusative subject, as in (50-b). Different from regular finite embedded clauses, we see that the 1st person indexical pronoun *beni* in the ASC is not targeted by indexical shift. To account for these facts, I propose that the indexical shifting OP_{AUTH} merges into the embedded clause of the ASCs without full, matching embedded verb agreement. This OP_{AUTH} is absent in ASCs that bear full, matching embedded verb agreement, as shown in (51).

- (51) If *Dursun* is speaking:

- a. Burak **ben-i** [CP Ankara-ya gel-di-m] san-ıyor.
 Burak 1SG-ACC Ankara-DAT come-PST-1SG think-PROG
 ‘Burak thinks that I came to Ankara.’ → ‘Dursun (or Addr) is in Ankara’
- b. Burak **ben-i** [CP OP_{AUTHi} Ankara-ya gel-di-Ø_i] san-ıyor.
 Burak 1SG-ACC Ankara-DAT come-PST think-PROG
 ‘Burak thinks that I went to Ankara.’ → ‘Burak is in Ankara’

All indexical elements under the scope of this OP_{AUTH} are shifted (e.g. *gel*), while the accusative-marked subject/agent stays unshifted since is out of the scope of the OP_{AUTH} . This is compatible with the generalizations made by previous work regarding the status of the accusative-marked subject/agent (e.g. Şener and Şener, 2011; Özyıldız et al., 2018).

4.2 Evidentiality

In this subsection I introduce new data with the evidential marker ‘-*mİş*’ inside embedded clauses of ASCs, demonstrating the correlation between embedded verb agreement and how the evidentiality is interpreted. When embedded verb carries matching/full 1st person agreement markers, the evidential markers have non-shifted interpretations. However, when embedded verb uses default (null) agreement, the evidential markers are shifted. Based on those facts, I conclude that indexical shifting OP merges ASCs without matching/full agreement and shifts the interpretations of evidential markers.

Data from the previous sections suggested that evidentiality in Turkish interacts with indexical shift, and the evidential marker ‘-*mİş*’ has shifted interpretations if it appears in an indexical shift environment. To capture the indexical-like behavior of ‘-*mİş*’, I suggested that its semantic definition should be roughly like (39-b), repeated here as (52).

(52) ‘-*mİş*’ = the *author* lacks direct evidence for reported proposition.

According to a definition like (52), ‘-*mİş*’ indicates that the speaker lacks direct evidence for the reported proposition in regular (non-shifted) environments. However, when it appears inside an environment where indexical shift took place, ‘-*mİş*’ should indicate that the attitude holder lacks evidence for the reported proposition, considering that indexical shift replaces the *author* parameter of the context with the attitude holder. Again, in light of this, consider the contrast between (53) and (54).

(53) *Dursun* is speaking:

- a. Burak **ben-i** uyu-du-m san-ıyor.
Burak 1SG-ACC sleep-PST-1SG think-PROG
‘Burak thinks that I fell asleep.’ → No information about why Burak thinks so
- b. #Burak **ben-i** uyu-muş-um san-ıyor.
Burak 1SG-ACC sleep-EVID-1SG think-PROG
Int.: ‘Burak thinks that I fell asleep.’ → ‘Dursun lacks evidence that Dursun was sleeping’

Both sentences in (53) carry embedded verb agreement that matches the features of the accusative subject (1st person), and they report the same embedded proposition with different past tense markers. (53-a) uses the regular past tense marker ‘-*DI*’ while (53-b) uses the evidential past tense marker ‘-*mİş*’. If uttered by *Dursun*, the regular past tense marker in (53-a) implies that *Dursun* has evidence for the reported situation (e.g. Dursun knows that he slept in the past, and now he is awake). The evidential marker in (53-b), on the other hand, implies that *Dursun* lacks evidence for the reported situation (e.g. Dursun lacks evidence that he slept), and is thus infelicitous. Examples in (53) show again that the evidentiality is evaluated based on the evidence that the speaker has for the reported proposition, as the default *author*. However, the evidential markers in ASCs with no (matching) embedded verb agreement are interpreted in a different way (54). It appears that in these sentences the evidentiality is interpreted based on the evidence or knowledge of the attitude holder (rather than the speaker).

(54) *Dursun* is speaking:

- a. Burak **ben-i** uyu-du-∅ san-ıyor.
Burak 1SG-ACC sleep-PST think-PROG
‘Burak thinks that I fell asleep.’ → Burak has strong evidence
- b. Burak **ben-i** uyu-muş-∅ san-ıyor.
Burak 1SG-ACC sleep-EVID think-PROG
‘Burak thinks that I fell asleep.’ → Burak has weak evidence (hear-say)

The pair of sentences in (54) lack the matching embedded verb agreement, and they differ in the evidentiality similar to the pair in (53). Interestingly, the evidentiality in both sentences is evaluated regarding the evidence that the attitude holder *Burak* has. More precisely, while the regular past tense marker in (54-a) implies that the attitude holder *Burak* has direct evidence for the reported situation (e.g. Burak saw Dursun [pretending to be] sleeping), the evidential marker in (54-b) implies that *Burak* has indirect evidence for the reported situation (e.g. Burak was told by others that Dursun fell asleep). Neither (54-a) or (54-b) has any implications regarding the evidence that the speaker has for the reported situation.

Considering that evidentials are evaluated based on the world knowledge of the *author*, this suggests that indexical shift occurred in embedded clauses of the sentences in (54), and the evidentiality is evaluated based on the evidence that the attitude holder *Burak* has, rather than the actual speaker *Dursun*.

The behavior observed here is very similar to that of *gel* ‘to come’ inside ASCs. Again the contrast between (53) and (54) suggests that indexical shift occurs in ASCs without embedded verb agreement (54) and forces the evidential marker to be interpreted in a context where the attitude holder is the *author*. To account for these facts, I suggest that the indexical shifting OP_{AUTH} merges with the embedded CP of ASCs that lack full, matching embedded verb agreement (55).

(55) *Dursun* is speaking:

- a. Burak **ben-i** [CP OP_{AUTHi} uyu-du- $\emptyset_i$] san-ıyor.
 Burak 1SG-ACC sleep-PST think-PROG
 ‘Burak thinks that I fell asleep.’ → Burak has strong evidence
- b. Burak **ben-i** [CP OP_{AUTHi} uyu-muş- $\emptyset_i$] san-ıyor.
 Burak 1SG-ACC sleep-EVID think-PROG
 ‘Burak thinks that I fell asleep.’ → Burak has weak evidence (hear-say)

Again, the accusative-marked subject/agent is not shifted since it is outside the scope of the OP_{AUTH} , while other indexical elements (e.g. evidential markers) are shifted.

4.3 Indexical pronouns embedded lower inside Accusative Subject Clauses (ASCs)

In previous subsections, I argued that indexical shift takes place in ASCs which lack embedded verb agreement. So far, the evidence for indexical shift in ASCs came from PSIs that are sensitive to indexical shift (i.e. *gel* ‘to come’, and evidential markers). In this subsection, I provide data with regular indexical elements like 1st person pronouns and show that their interpretations are correlated with the choice of the embedded verb agreement. Similar to previous patterns, we observe that when the embedded verb carries full/matching 1st person agreement, 1st person indexicals stay non-shifted. However, when embedded verb uses default (null) agreement, 1st person indexicals have shifted interpretations.

As discussed in previous sections, the accusative-marked subject/agent is predicted to stay non-shifted even if the OP merges the structure (e.g. Şener and Şener, 2011). So, in this subsection I use ASC examples where a 1st person indexical appears in a position other than the ‘embedded subject’ of the embedded clause. For example, in (56), the 1st person *ben* ‘I’ is in the embedded object position of the ASC, and thus is marked with accusative case.

- (56) Icardi **Wanda-yı** hiçbir zaman **ben-i** aldat-maz- $\emptyset$ san-ıyor-du.
 Icardi Wanda-ACC any time 1SG-ACC cheat-NEG-3SG think-PROG-PST
 ‘Icardi thought that Wanda would never cheat on {me/him}’.

Crucially, 1st person the indexical *ben-i* can be interpreted as referring to *the speaker* (non-shifted) or the attitude holder *Icardi* (shifted). In (56), the accusative subject is a 3rd person entity (i.e. *Wanda*), so 1st person agreement is not available on the embedded verb. Here, we see that the sentence is simply ambiguous between shifted vs non-shifted readings of the 1st person indexical in the embedded object position.

Moreover, sentences in (57) demonstrate that the occurrence of indexical shift is reflected by embedded verb agreement. In (57), the accusative subject is the 1st person indexical, and there is another 1st person indexical in the embedded (indirect) object position, marked with ablative case.

- (57) a. #Meryem **ben-i** üniversite-den mezun ol-unca **ben-den** ayrıl-acağ-ım
 Meryem 1SG-ACC university-ABL graduate become-WHEN 1SG-ABL break.up-FUT-1SG
 san-ıyor-du.
 think-PROG-PST
 Lit: ‘Meryem thought that I would break up with **me** after graduating from the University.’

- b. Meryem **ben-i** üniversite-den mezun ol-unca **ben-den** ayrıl-acak- $\emptyset$
 Meryem 1SG-ACC university-ABL graduate become-WHEN 1SG-ABL break.up-FUT-3SG
 san-ıyor-du.
 think-PROG-PST
 ‘Meryem thought that I would break up with **her** after graduating from the University.’

As translations indicate, the sentences (57-a) and (57-b) have noticeably different meanings. Though the 1st person accusative subject (*ben-i*) can only refer to the speaker, as discussed in earlier subsections, the interpretation of the embedded object indexical (*ben-den*) changes depending on the embedded verb agreement. Precisely, the embedded object indexical (*ben-den*) stays non-shifted when the embedded verb carries full, matching agreement marker (57-a), but it shifts when the agreement marker is missing (57-b). This particular example is semantically odd when indexical shift does not occur, embedded verb meaning something like ‘... I would break up with me after graduation...’, so the example is not acceptable with the full agreement marker on the embedded verb. This highlights the semantic contributions of embedded verb agreement in ASCs, demonstrating that embedded verb agreement is not totally optional or random.

Another piece of evidence for indexical shift in ASCs comes from possessive pronouns. Importantly, Turkish requires possessive agreement morphology on the possessed noun, matching the features of the possessor DP. This is illustrated in (58), where possessed noun *anne* ‘mother’ carries either 1st person singular (58-a) or 1st person plural agreement (58-c). Moreover, Turkish allows (and usually prefers) the possessor DP to be silent (dropped), as shown in (58-b) and (58-d). In such cases, agreement markers on the possessed noun indicates the features of the possessor DP.

- (58) a. **Benim** anne-**m** gel-di. c. **Bizim** anne-**miz** gel-di.
 1SG.GEN mother-1SG come-PST 1PL.GEN mother-1PL come-PST
 ‘My mother came.’ ‘My mother came.’
 b. *pro*.1SG anne-**m** gel-di. d. *pro*.1PL anne-**miz** gel-di.
 1SG.GEN mother-1SG come-PST 1PL.GEN mother-1PL come-PST
 ‘My mother came.’ ‘My mother came.’

Sentences like (59) show that the interpretation of the 1st person possessive pronouns (silent, *pro*) inside the embedded clauses of ASCs is affected by the optionality of embedded agreement.

- (59) a. Wanda_i **ben-i_k** evlen-ince *pro*.1SG_k çocuk-lar-ım-a bak-ar-ım
 Wanda 1SG-ACC marry-WHEN 1SG-GEN child-PLU-1SG-DAT care-AOR-1SG
 san-ıyor-du.
 think-PROG-PST
 ‘Wanda_i thought that I_k would take care of **my_k** children after getting married.’
 b. Wanda_i **ben-i_k** evlen-ince *pro*.1SG_i çocuk-lar-ım-a bak-ar- $\emptyset$
 Wanda 1SG-ACC marry-WHEN 1SG-GEN child-PLU-1SG-DAT care-AOR-3SG
 san-ıyor-du.
 think-PROG-PST
 ‘Wanda_i thought that I_k would take care of **her_i** children after getting married.’

In (59), the (silent) 1st person possessor stays non-shifted and is interpreted as referring to the speaker (i.e. ‘Speaker takes care of speaker’s children’) when the embedded verb carries full, matching agreement with the accusative subject (59-a), but it shifts and is interpreted as referring to the attitude holder *Wanda* (i.e. ‘Speaker takes care of Wanda’s children’) when embedded verb lacks this agreement marker (59-b). The fact that the 1st person possessor DP is interpreted as referring to *Wanda* in (59-b) suggests that the *author* of the embedded clause in (59-b) is *Wanda*, rather than the actual speaker. This is in line with data from previous subsections, where we saw that the lack of embedded verb agreement in ASCs signaled that indexical shift took place in the embedded clause and the indexical elements inside the embedded clause domain were interpreted in shifted readings.

Moreover, Turkish also allows sentences like (60), where a non-indexical pronoun (e.g. 3rd person) gets shifted interpretations, depending on the embedded verb agreement. Here, the possessed noun carries 3rd person agreement marker, implying that the possessor is a 3rd person (silent) pronoun.

- (60) a. Wanda_i **ben-i**_k evlen-ince *pro*.1SG_k çocuk-lar-ı-na bak-ar-ım
 Wanda 1SG-ACC marry-WHEN 1SG-GEN child-PLU-3SG-DAT care-AOR-1SG
 san-ıyor-du.
 think-PROG-PST
 ‘Wanda_i thought that I_k would take care of **her**_i children after getting married.’
 b. Wanda_i **ben-i**_k evlen-ince *pro*.1SG_i çocuk-lar-ı-na bak-ar-∅
 Wanda 1SG-ACC marry-WHEN 1SG-GEN child-PLU-3SG-DAT care-AOR-3SG
 san-ıyor-du.
 think-PROG-PST
 ‘Wanda_i thought that I_k would take care of **my**_k children after getting married.’

In (60-a), with the embedded verb agreement, the (silent) 3rd person possessive pronoun is interpreted as referring to *Wanda* (or any other salient 3rd person), leading to a reading like ‘Speaker takes care of Wanda’s children’. In other words, as expected, the 1st person pronoun refers to the speaker, while the 3rd person possessor refers to the salient antecedent *Wanda*. In (60-b), on the other hand, both the 1st person accusative pronoun *ben-i* and the 3rd person possessor are interpreted as referring to the speaker, leading to an interpretation like ‘Speaker takes care of speaker’s children’. The crucial observation here is that in ASCs with no embedded verb agreement, such as (60-b), the actual speaker can use a 3rd person pronoun inside the embedded clause to refer to themselves, but they cannot use a 1st person pronoun, which would refer to the attitude holder as in (60-a).

In short, we examine how indexical pronominals behave inside ASCs in positions other than the so-called ‘embedded subject’ position. Through the examples (56)-(60), we observe that the interpretation of 1st person and 3rd person pronominal change depending on the embedded verb agreement in ASCs. More precisely, when the embedded verb lacks matching agreement with the accusative subject, 1st person elements are interpreted as referring to the attitude holder and the 3rd person elements are interpreted as referring to the speaker, signaling indexical shift.

To account for these facts, I argue below that indexical shift takes place in the embedded clauses of ASCs with no embedded verb agreement, and requires any 1st person elements in the embedded clause (e.g. embedded objects, possessor DPs) to be interpreted as referring to the attitude holder. When indexical shift takes place, the speaker cannot refer to themselves with a 1st person pronominal anymore since they are not the *author* of the embedded clause, and thus they must use a 3rd person pronoun to refer to themselves as in (60-b).

4.4 Final piece of evidence for Indexical Shift in ASCs: Locative indexicals

This subsection provides one final piece of evidence for indexical shift in ASCs, using independent evidence regarding how different indexical elements behave under different embedding verbs. Indexical shift in Turkish targets different indexical elements under different embedding verbs (e.g. Özyıldız, 2012; Oğuz et al., 2020). For example, some indexicals like *bura* ‘here’ can shift under the embedding verb *de* ‘to say’, but they cannot shift under the embedding verb *san* ‘to think’. This is illustrated in (61), repeated from (18).

- (61) If uttered by *Dursun* in *Los Angeles*:
 a. Ali İzmir-de [**bura**-nın ne-ler-i güzel] de-miş-ti?
 Ali Izmir-LOC here-GEN what-PLU-3SG.POSS pretty say-DUB-PST
 ‘In Izmir, what did Ali say was pretty {Izmir/Los Angeles}?’
 b. Ali İzmir-de [**bura**-nın ne-ler-i güzel] san-mış-tı?
 Ali Izmir-LOC here-GEN what-PLU-3SG.POSS pretty think-DUB-PST
 ‘In Izmir, what did Ali think was pretty {*Izmir/Los Angeles}?’

The indexical *bura* ‘here’ in (61-a) is ambiguous between shifted and non-shifted readings under the embedding verb *de*, while (61-b) shows that *bura* ‘here’ can only have non-shifted interpretations with the embedding verb *san*. This contrast between *de* ‘to say’ and *san* ‘to think’ is reported in previous work (e.g. Özyıldız, 2012; Oğuz et al., 2020), and is in line with the cross-linguistic patterns of indexical shift, described in Deal (2020).

To account for this difference between *de* ‘to say’ and *san* ‘to think’, Oğuz et al. (2020) adopt the shifty operator analysis developed by Deal (2020) and argue that each context parameter can be targeted by the merge of a separate shifty operator in the left periphery of finite embedded clauses in Turkish. For example, the 1st person indexicals shift due to the merge of OP_{AUTH} , while the locative indexical *bura* ‘here’ shifts due to the merge of a separate operator, namely OP_{LOC} . Under this view, an embedding verb like *de* ‘to say’ can select for an embedded CP with both OP_{AUTH} and OP_{LOC} and thus can shift both types of indexicals, while an embedding verb like *san* ‘to think’ can select for embedded CPs with OP_{AUTH} but not OP_{LOC} and as a result cannot shift the locative indexicals like *bura* ‘here’.

So, if the shifty behavior we observe in ASCs is a true instance of indexical shift, this would predict that the locative indexical *bura* ‘here’ in ASCs should not be able to shift under the embedding verb *san* ‘to think’, but it should be able to shift under the embedding verb *de* ‘to say’. This prediction is borne out, as shown in (62)¹².

(62) If uttered by *Dursun* in *Los Angeles*:

- a. Ali İzmir-de-yken İsa-yı **bura**-da doğ-du- $\emptyset$ de-miş-ti.
 Ali İzmir-LOC-WHEN Jesus-ACC here-LOC be.born-PST say-DUB-PST
 ‘When in İzmir, Ali said that Jesus was born in {İzmir/Los Angeles}.’
- b. Ali İzmir-de-yken İsa-yı **bura**-da doğ-du- $\emptyset$ san-mış-tı.
 Ali İzmir-LOC-WHEN Jesus-ACC here-LOC be.born-PST think-DUB-PST
 ‘When in İzmir, Ali thought that Jesus was born in {*İzmir/Los Angeles}.’

Bura ‘here’ in (62-a) is ambiguous between the shifted (*İzmir*) and non-shifted (*Los Angeles*) readings, while *bura* ‘here’ in (62-b) can only be interpreted in the non-shifted reading (*Los Angeles*). This suggests that the shifty behavior observed with the ASCs is the same indexical shifting operation that is observed with regular finite embedded clauses, and the embedding verb *san* ‘to think’ cannot shift the indexical *bura* ‘here’, in line with findings of the previous studies (e.g. Özyıldız, 2012; Oğuz et al., 2020).

4.5 Extending the analysis to Accusative Subject Clauses (ASCs)

In summary, though Accusative Subject Clauses (ASCs) in Turkish have been claimed to not allow indexical shift based on the unavailability of the accusative-marked subject to shift, a closer investigation with novel data demonstrates that indexical shift can optionally take place in ASCs and affect how indexical elements, including indexical PSIs, are interpreted inside the embedded clauses of ASCs. Moreover, novel data introduced in this paper shows that the embedded verb agreement is not totally optional or random, but it indicates whether indexical shift takes place or not. More precisely, the lack of (full, matching) agreement on the embedded verb indicates that indexical shift occurs in the embedded clause and that indexical elements in this domain are interpreted in a context where the *author* is the attitude holder. When the embedded verb agrees with the features of the accusative subject, it indicates that indexical shift has not happened and the indexical elements are interpreted from the actual speech context, where the *author* is the speaker (by default). This contrasts previous work on the topic, which assumed that the embedded verb agreement was purely syntactic and did not affect the meaning of ASCs (e.g. Şener, 2008; Özgen and Aydın, 2016; Başoğlu, 2021).

To account for these facts, I follow Oğuz et al. (2020) and adopt the shifty operator analysis developed by Deal (2020) to suggest that each context parameter can be targeted by the merge of a separate shifty operator in the left periphery of finite embedded clauses in Turkish. The merge of these operators shift the context in which the indexical items (including Indexical PSIs) under their scopes are interpreted, and lead to shifted readings of the relevant indexical elements. Our data with locative indexical *bura*

¹² It is worth noting that some speakers (around 50% of our informants) do not accept *de* ‘to say’ as an embedding verb for Accusative Subject Clauses (ASCs). The judgments of (62-a) come from those speakers who accept *de* ‘to say’ as an embedding verb for ASCs.

‘here’ shows that locative indexicals cannot shift under the embedding verb *san* ‘to think’, while they can shift under the embedding verb *de* ‘to say’ (in line with data from Özyıldız (2012) and Oğuz et al. (2020)). This suggests that two operators, namely OP_{AUTH} and OP_{LOC} merge in the left periphery of embedded CPs under the verb *de* ‘to say’, while only OP_{AUTH} merges in the embedded CPs under the verb *san* ‘to think’.

(63) If uttered by *Dursun* in *Los Angeles*:

- a. Ali İzmir-de-yken İsa-yı [CP (OP_{LOC_i}) (OP_{AUTH}) **bura_i**-da doğ-du- $\emptyset$]
 Ali İzmir-LOC-WHEN Jesus-ACC here-LOC be.born-PST
 de-miş-ti.
 say-DUB-PST
 ‘When in İzmir, Ali said that Jesus was born in {İzmir/Los Angeles}.’
- b. Ali İzmir-de-yken İsa-yı [CP (OP_{AUTH}) **bura**-da doğ-du- $\emptyset$] san-mış-tı.
 Ali İzmir-LOC-WHEN Jesus-ACC here-LOC be.born-PST think-DUB-PST
 ‘When in İzmir, Ali thought that Jesus was born in {*İzmir/Los Angeles}.’

As a novel contribution, I suggest these operators can merge into finite embedded CPs, regardless of the case of its subject (nominative vs accusative subject clauses). In ASCs, the lack of full/matching embedded verb agreement with the accusative-marked subject/agent indicates the merge of the indexical shifting operator(s).

One important observation to account for is that the accusative subject in ASCs cannot have shifted interpretations even when other indexical elements in the embedded clause undergo indexical shift. I suggest that this is due to the syntactic position of the accusative subject DP (DP_{ACC}). Previous research suggests that the DP_{ACC} merges in the embedded subject position but then raises higher into the matrix object position (e.g. Zidani-Eroğlu, 1997) or into the topic position in the left periphery of the embedded CP (e.g. Şener, 2008). A closely parallel structure in Uyghur is analyzed to be proleptic at least in some cases, where DP_{ACC} merges as the object of the matrix verb and is in coreference with another pronominal inside the embedded clause (Major, 2022). Across all of these analyses, the DP_{ACC} occupies a position (at least raises into) a syntactic position higher than the indexical shifting operators, and thus escapes their scopes. As a result, the analysis developed in this study is compatible with all of these analyses.

Though the current study stays neutral regarding the syntactic status of ASCs in Turkish, the proleptic analysis is the most compatible with the analysis developed here, considering Turkic and other cross-linguistic tendencies. First of all indexical shift (the merge of the OP) in ASCs is unexpected under the ECM/SOR analyses, considering that indexical shifting OP typically merges finite embedded clauses with full TP/CP layers (e.g. Deal, 2020), and ECM (e.g. Şener, 2008) and SOR structures (e.g. Zidani-Eroğlu, 1997) are argued to lack these layers. If these standard analyses of ECM or SOR are adopted, this might require adjustments to the shifty OP analyses of indexical shift (e.g. Deal, 2020), allowing the operators to merge in non-finite embedded CPs as well. Moreover, Major (2022) investigates indexical shift in Uyghur, a closely related Turkic language, and suggests that indexical shifting OP can only merge in finite clauses that bear full verb agreement (but not ECM/SOR structures), confirming the cross-linguistic predictions of Deal (2020). In his later work, Major (2024) adds that Uyghur has proleptic structures that look almost identical to ASCs (except for embedded verb agreement). Considering these, it is possible that at least some Turkish ASCs are indeed proleptic structures. If Turkish ASCs with indexical shift are analyzed as proleptic structures, Turkish would be no exception to Turkic and cross-linguistic patterns regarding indexical shift.

Under a proleptic analysis, the embedded verb in ASCs would not be defective (as in ECM/SOR analyses) but would always be projected. Under such an analysis, the apparent lack of agreement marker sentences like (50-b) could be analyzed as 3rd person agreement, which is morphologically null in Turkish. (50-b) would schematically look like (64) under a proleptic analysis.

- (64) Burak **ben-i_i** [CP OP_{AUTH} *pro*.3SG_i Ankara-ya gel-di- $\emptyset$] san-ıyor.
 Burak 1SG.ACC Ankara-DAT come-PST-3SG think-PROG
 ‘Burak thinks of me_i *asthathe_icametoAnkara.*’ → ‘Burak is in Ankara’

In a proleptic structure like (64), the 1st person accusative subject/agent functions like a hanging topic of the embedded clause that introduces an aboutness relation to be satisfied in the embedded clause, and thus a pronominal inside the embedded clause should refer back to the accusative subject. However, since the indexical shifting OP merges into the embedded CP layer, the *author* inside the embedded clause is not the actual speaker but is the attitude holder *Burak*. As a result, a 1st person pronoun cannot be used to refer to the actual speaker (as also demonstrated in (59) and (60)). Instead, a 3rd person (silent) pronoun is used, which triggers 3rd person embedded verb agreement (morphologically null).

However, if ASCs are analyzed as SOR structures, the accusative-marked DP would first merge into the embedded subject position, and then raise higher into the matrix clause domain, escaping the scope of the OP. (50-b) would schematically look like (65) under the SOR analysis. Crucially in the SOR analysis, the embedded verb agreement is not projected at all, due to defective T/C heads in the embedded clause.

- (65) Burak **ben-i_i** [CP/TP OP_{AUTH} [*t_i*Ankara-ya gel-di] san-ıyor.
 Burak 1SG.ACC Ankara-DAT come-PST think-PROG
 ‘Burak thinks of me_i*asthathe_icametoAnkara.*’ → ‘Burak is in Ankara’

Alternatively, if we analyze ASCs as ECM structures, the accusative-marked DP merges into the embedded subject position and raises higher into the periphery of the embedded CP (but not as high as into matrix clause domain). The fact that the accusative DP cannot shift when indexical shift occurs suggests that it raises into a position higher than the OP (e.g. topic position) and escapes its scope. So, (50-b) would schematically look like (66) under the ECM accounts of ASCs. Again, according to the ECM analysis, the embedded verb agreement is not projected due to defective T/C heads in the embedded clause.

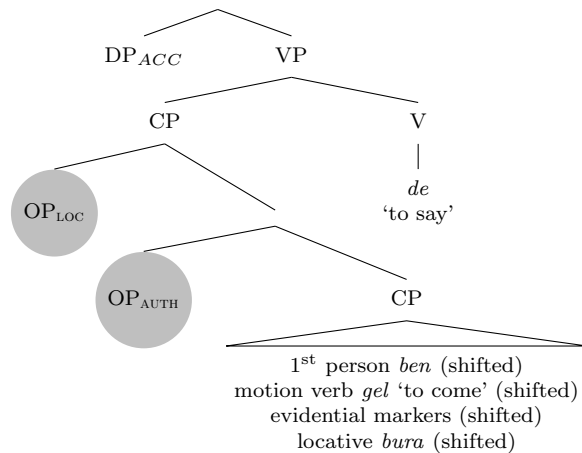
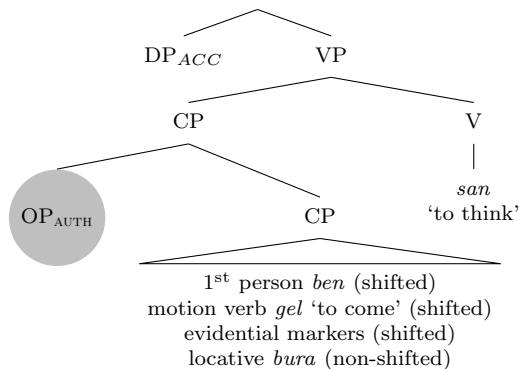
- (66) Burak [CP **ben-i_i** [CP/TP OP_{AUTH} [*t_i*Ankara-ya gel-di] san-ıyor.
 Burak 1SG.ACC Ankara-DAT come-PST think-PROG
 ‘Burak thinks of me_i*asthathe_icametoAnkara.*’ → ‘Burak is in Ankara’

Even though the analysis developed in this paper does not rely on any specific syntactic analysis of ASCs, the indexical shift facts can more straightforwardly be accounted for if ASCs with indexical shift are analyzed as proleptic structures as explained above. If ECM/SOR analyses are adopted, one would need to argue that these are exceptional cases where an indexical shifting OP can merge a non-finite CP, contrasting other Turkic and cross-linguistic patterns.

Trees in (67) illustrate these proposed structures, under a proleptic analysis where accusative-marked subject/agent merges as the matrix object and is in co-reference with a pronominal in the embedded clause¹³.

¹³ The tree representations for the SOR and ECM analyses are not provided due to space limitations. Under the SOR analyses, the DP_{ACC} originates in the embedded clause and raises into its final position. Under the ECM analysis, the DP_{ACC} originates in the embedded clause and raises into the left periphery of the embedded CP.

(67)

a. *Shifty OPs embedded under de* ‘to say’b. *Shifty OPs embedded under san* ‘to think’

Lastly, it is worth mentioning that the existence of indexical PSIs suggests that elements not typically viewed as indexical (e.g. motion verb *gel*) can exhibit indexical properties and may be targeted by the same kind of shifting operator. This supports context-overwriting approaches to indexical shift (e.g. Anand and Nevins, 2004; Deal, 2020) and is not directly predicted by approaches that reduce indexical shift to syntactic agreement and pronominal feature/index manipulation (e.g. Podobryaev, 2014; Messick, 2023). Though the details of these accounts are different, they rely on the manipulation of complex indices/features of pronominal elements in syntax. However, considering that indexical PSIs (e.g. motion verb *gel* ‘to come’ and evidential markers) are not pronominal elements, it is not directly clear how these approaches to indexical shift could account for the existence of indexical PSIs. On a shifty OP analysis on the other hand, the indexical shifting OP can target all elements that have indexical properties of the relevant type (e.g. *author*), so the shifting (and shift together) facts of the indexical PSIs are more straightforwardly explained. An alternative account has been proposed by Baker (2023), who argues that indexical shift is a result of syntactic binding/control relation between syntactic *Speaker/Adressee* heads and indexical pronominals. Though indexical PSIs do not provide any clear evidence against Baker’s approach, it faces some questions regarding how and why separate indexical elements behave differently under different embedding verbs, thus the current study concludes that the data introduced here can be best accounted for under the shifty operator accounts (e.g. Anand and Nevins, 2004; Deal, 2020).

5 Conclusion

In summary, this study provides novel data relevant for how perspective sensitive elements (PSIs) in environments where indexical shift occurs. The study has two main findings. First, I show that different types of PSIs show different properties. While most types of PSIs are not affected by indexical shift, a subtype of PSIs like motion verb *gel* ‘to come’ and evidential markers, which I call *indexical PSIs*, behave like indexicals and show shift together patterns with 1st person indexicals (e.g. they either shift together or neither of them shift). To account for the facts observed here, I propose that the definitions of these *indexical PSIs* must refer to the *author* parameter of the context, and thus have shifted interpretations whenever indexical shift takes place, as context-overwriting accounts of indexical shift predict (e.g. Anand and Nevins, 2004; Deal, 2020). Second, using novel data provided in this study, I show that indexical shift can take place in Turkish Accusative Subject Clauses (ASCs), which have previously been reported to disallow indexical shift. The careful investigation of Turkish data showed that the optionality of the agreement marker on the embedded verb of ASCs is not random or coincidental, but it indicates whether perspective shift took place in the embedded clause or not.

To account for these facts, I use the shifty operators analysis of Deal (2020), adopted for Turkish by Oğuz et al. (2020), and suggest that context-overwriting operators (e.g. OP_{AUTH}) can merge in the left periphery of finite embedded clauses in Turkish (including ASCs) and replace the *author* parameter of the context in which the sentence is interpreted in. For the PSIs that do not exhibit indexical-like properties, I follow

existing literature which suggests that PSIs are interpreted relative to another coordinate of the context such as *judge* (e.g. Lasersohn, 2005).

The findings of the current study has implications for theoretical approaches to indexical shift, and also for the syntactic work on Turkish/Turkic. Firstly, the existence of indexical PSIs provide support for the context-overwriting approaches to indexical shift (e.g. Anand and Nevins, 2004; Deal, 2020) over the syntactic binding/control accounts (e.g. Podobryaev, 2014; Messick, 2023). Secondly, availability of indexical shift in ASCs implies that at least some instances of ASCs could be analyzed as proleptic structures with full, finite embedded CPs. Considering that Turkish ECM/SOR structures have been used in different arguments for cross-clausal operations and features of CP (e.g. Özgen and Aydın, 2016; Wurmbrand, 2019), if the prolepsis analysis proposed here is on the right track, it would potentially have important implications for the literature regarding the literature and might suggest reconsideration of arguments put forward in recent years.

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